

# SDS $_{class}$ Notes

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# Contents

|          |                                                   |           |
|----------|---------------------------------------------------|-----------|
| <b>1</b> | <b>Introduction: 26/09/2024</b>                   | <b>6</b>  |
| 1.1      | Sample Space . . . . .                            | 6         |
| 1.1.1    | Union of Sets . . . . .                           | 6         |
| 1.1.2    | Intersection of Sets . . . . .                    | 6         |
| 1.1.3    | Complement of a Set . . . . .                     | 6         |
| 1.1.4    | Venn Diagrams . . . . .                           | 6         |
| 1.2      | De Morgan's Laws . . . . .                        | 7         |
| 1.3      | Elementary Probability . . . . .                  | 7         |
| 1.3.1    | Classical Definition . . . . .                    | 7         |
| 1.3.2    | Frequentist Definition . . . . .                  | 8         |
| 1.3.3    | Axiomatic Definition (Take 1) . . . . .           | 8         |
| 1.4      | Inclusion-Exclusion Principle . . . . .           | 8         |
| 1.4.1    | Two Events . . . . .                              | 8         |
| 1.4.2    | Three Events . . . . .                            | 8         |
| 1.5      | Boole's Inequality (a.k.a. Union Bound) . . . . . | 9         |
| 1.6      | Bonferroni's Inequality . . . . .                 | 9         |
| 1.7      | Conditional Probability . . . . .                 | 10        |
| 1.8      | Multiplication Rule . . . . .                     | 10        |
| 1.9      | Chain Rule . . . . .                              | 10        |
| 1.10     | Independence . . . . .                            | 11        |
| 1.11     | The Law of Total Probabilities . . . . .          | 11        |
| 1.12     | Bayes' Theorem . . . . .                          | 12        |
| 1.12.1   | Probability and Statistics . . . . .              | 12        |
| <b>2</b> | <b>Random Variables:01/10/2024(4a)</b>            | <b>13</b> |
| 2.1      | Remark . . . . .                                  | 13        |
| 2.2      | Random Variables and Distributions . . . . .      | 13        |
| 2.3      | Basic Properties . . . . .                        | 14        |
| 2.3.1    | Basic Properties of PMF . . . . .                 | 14        |
| 2.3.2    | Basic Properties of PDF . . . . .                 | 14        |
| 2.3.3    | Basic Properties of CDF . . . . .                 | 15        |
| 2.4      | Example: Fair Dice (Discrete Case) . . . . .      | 16        |
| 2.5      | The Quantile Function . . . . .                   | 17        |
| <b>3</b> | <b>Univariate Distributions(4b)</b>               | <b>18</b> |
| 3.1      | Bernoulli Distribution . . . . .                  | 18        |
| 3.2      | Binomial Distribution . . . . .                   | 18        |
| 3.3      | Geometric Distribution . . . . .                  | 19        |

|          |                                                              |           |
|----------|--------------------------------------------------------------|-----------|
| 3.4      | Poisson Distribution . . . . .                               | 20        |
| 3.5      | Uniform Distribution . . . . .                               | 20        |
| 3.6      | Exponential Distribution . . . . .                           | 21        |
| 3.7      | Normal/Gaussian Distribution . . . . .                       | 22        |
| 3.8      | Student's t-Distribution . . . . .                           | 22        |
| 3.9      | Laplace Distribution . . . . .                               | 23        |
| 3.10     | Comparison between Normal, T-student and Laplacian . . . . . | 24        |
| 3.11     | Gamma Distribution . . . . .                                 | 24        |
| 3.12     | Beta Distribution . . . . .                                  | 25        |
| 3.13     | Exponential Family . . . . .                                 | 25        |
| <b>4</b> | <b>Fam.Prob - Models: 04/10/2024</b>                         | <b>26</b> |
| 4.0.1    | Learning Tasks and Loss Functions . . . . .                  | 26        |
| 4.1      | Logistic Regression Example . . . . .                        | 26        |
| 4.1.1    | Modeling Logistic Regression . . . . .                       | 26        |
| 4.1.2    | Link Function and Logistic Function . . . . .                | 27        |
| 4.2      | Sampling and Likelihood . . . . .                            | 27        |
| 4.2.1    | Maximizing the Likelihood . . . . .                          | 27        |
| 4.3      | Random Vectors . . . . .                                     | 27        |
| 4.3.1    | Case 1: All Discrete Random Variables . . . . .              | 27        |
| 4.3.2    | Joint PMF: Topic 09/10 . . . . .                             | 28        |
| 4.3.3    | Bivariate CDF . . . . .                                      | 29        |
| 4.3.4    | Joint PDF (Continuous Case) . . . . .                        | 29        |
| 4.3.5    | Example . . . . .                                            | 30        |
| 4.4      | Histogram and ECDF: 11/10 . . . . .                          | 32        |
| 4.4.1    | Histogram . . . . .                                          | 32        |
| 4.4.2    | Empirical Cumulative Distribution Function (ECDF) . . . . .  | 33        |
| 4.5      | Mixed Case: Lesson 16/10 . . . . .                           | 33        |
| 4.5.1    | Gaussian Mixture Models (GMM) . . . . .                      | 34        |
| 4.5.2    | Scale Mixture of Gaussians . . . . .                         | 35        |
| 4.5.3    | Discrete Observations + Cont' Latent Variables . . . . .     | 36        |
| <b>5</b> | <b>Expectation e Variance: 5</b>                             | <b>37</b> |
| 5.1      | Introduction to Expectation: 23/10 . . . . .                 | 37        |
| 5.1.1    | Properties of Expectation . . . . .                          | 37        |
| 5.1.2    | Prediction under Perfect Information . . . . .               | 40        |
| 5.2      | Variance and Its Properties: 25/10 . . . . .                 | 40        |
| 5.2.1    | Law of Large Numbers (LLN) . . . . .                         | 43        |
| 5.2.2    | Conditional Expectation . . . . .                            | 43        |
| 5.2.3    | Random Vector $(X, Y)$ . . . . .                             | 44        |
| 5.2.4    | Law of Total Expectation . . . . .                           | 45        |
| 5.2.5    | Law of Total Variance . . . . .                              | 45        |
| <b>6</b> | <b>Dependency and correlation</b>                            | <b>46</b> |
| 6.1      | Overview . . . . .                                           | 46        |
| 6.2      | Linear Dependency . . . . .                                  | 46        |
| 6.2.1    | Standardization and correlation . . . . .                    | 47        |
| 6.2.2    | Geometry of Correlation . . . . .                            | 47        |
| 6.2.3    | Notation . . . . .                                           | 48        |

|           |                                                           |           |
|-----------|-----------------------------------------------------------|-----------|
| <b>7</b>  | <b>Graphical Models [7b]</b>                              | <b>49</b> |
| 7.0.1     | Motivation . . . . .                                      | 49        |
| 7.0.2     | Conditional Independence and Factorization . . . . .      | 49        |
| 7.0.3     | Types of Graphical Models . . . . .                       | 49        |
| 7.1       | Directed Graphical Models (Bayesian Networks) . . . . .   | 50        |
| 7.1.1     | Terminology . . . . .                                     | 50        |
| 7.1.2     | Markov Condition in Bayesian Networks . . . . .           | 50        |
| 7.1.3     | Graph Induced Factorization . . . . .                     | 51        |
| 7.2       | d-Separation . . . . .                                    | 51        |
| <b>8</b>  | <b>Probability Inequalities [10]</b>                      | <b>53</b> |
| 8.1       | Convergence . . . . .                                     | 55        |
| 8.1.1     | Convergence in Probability . . . . .                      | 55        |
| 8.1.2     | Almost Sure Convergence . . . . .                         | 55        |
| 8.1.3     | Laws of Large Numbers (LLN) . . . . .                     | 55        |
| 8.1.4     | The Central Limit Theorem (CLT) . . . . .                 | 56        |
| <b>9</b>  | <b>Statistical Inference [11]</b>                         | <b>57</b> |
| 9.1       | Statistical Inference . . . . .                           | 57        |
| 9.1.1     | Inferential Paradigms: Frequentist vs. Bayesian . . . . . | 57        |
| 9.1.2     | Parametric vs. Nonparametric Inference . . . . .          | 58        |
| 9.1.3     | Main Learning Problems in Statistical Inference . . . . . | 59        |
| 9.2       | Point Estimation . . . . .                                | 59        |
| 9.2.1     | Risk Function and Expectation of Loss . . . . .           | 60        |
| 9.2.2     | Consistency and Asymptotics . . . . .                     | 60        |
| 9.3       | Nonparametric Point Estimation . . . . .                  | 61        |
| 9.3.1     | Empirical Distribution Function (EDF) . . . . .           | 61        |
| 9.3.2     | Uniform Estimation (Estimating the Entire CDF) . . . . .  | 62        |
| 9.4       | Introduction to Set Estimation . . . . .                  | 63        |
| 9.4.1     | Confidence Intervals . . . . .                            | 63        |
| 9.5       | Statistical Functionals . . . . .                         | 64        |
| 9.5.1     | Plug-in Principle . . . . .                               | 64        |
| <b>10</b> | <b>Bootstrap</b>                                          | <b>66</b> |
| 10.0.1    | Bootstrap Resampling Procedure . . . . .                  | 66        |
| 10.1      | Bootstrapping Standard Errors . . . . .                   | 67        |
| 10.1.1    | Monte Carlo Approximation way . . . . .                   | 67        |
| 10.2      | Bootstrap Confidence Intervals . . . . .                  | 68        |
| 10.2.1    | Normal Approximation Interval . . . . .                   | 68        |
| 10.2.2    | Percentile Bootstrap Interval . . . . .                   | 68        |
| 10.2.3    | Pivotal Bootstrap Interval . . . . .                      | 68        |
| <b>11</b> | <b>Parametric Inference and the Parametric Bootstrap</b>  | <b>69</b> |
| 11.0.1    | Parametric Inference . . . . .                            | 69        |
| 11.1      | Method of Moments Estimation . . . . .                    | 69        |
| 11.2      | Maximum Likelihood Estimation (MLE) . . . . .             | 69        |
| 11.3      | Properties of MLE . . . . .                               | 70        |
| 11.4      | Maximum Likelihood Estimators . . . . .                   | 70        |
| 11.4.1    | Asymptotic Normality of MLE . . . . .                     | 71        |

|           |                                                            |           |
|-----------|------------------------------------------------------------|-----------|
| 11.4.2    | Example: Bernoulli Model . . . . .                         | 71        |
| 11.4.3    | Normal Confidence Intervals . . . . .                      | 72        |
| 11.5      | Normal Intervals for the Mean (Asymptotic Pivot) . . . . . | 72        |
| 11.5.1    | The $\delta$ -Method . . . . .                             | 73        |
| 11.6      | Parametric Bootstrap . . . . .                             | 73        |
| 11.6.1    | Nonparametric Bootstrap . . . . .                          | 74        |
| 11.6.2    | Parametric Bootstrap . . . . .                             | 74        |
| 11.6.3    | Importance of Model Assumptions . . . . .                  | 74        |
| 11.7      | Model Criticism and Selection . . . . .                    | 75        |
| 11.7.1    | How Do We Choose the Right Model? . . . . .                | 75        |
| 11.7.2    | Example: Modeling Failure Times . . . . .                  | 75        |
| 11.8      | Akaike Information Criterion (AIC) . . . . .               | 75        |
| 11.8.1    | AIC: Definition and Motivation . . . . .                   | 75        |
| 11.8.2    | Justification via Kullback-Leibler Divergence . . . . .    | 76        |
| 11.9      | Cross-Validation (CV) . . . . .                            | 76        |
| 11.9.1    | Basic Idea . . . . .                                       | 76        |
| 11.9.2    | Single-Split Cross-Validation . . . . .                    | 76        |
| 11.10     | Bayesian Information Criterion (BIC) . . . . .             | 76        |
| 11.10.1   | Definition and Interpretation . . . . .                    | 76        |
| 11.10.2   | Bayesian Justification . . . . .                           | 77        |
| 11.11     | Model Selection . . . . .                                  | 77        |
| 11.11.1   | Akaike Information Criterion (AIC) . . . . .               | 77        |
| 11.11.2   | Cross-Validation for Model Selection . . . . .             | 78        |
| 11.11.3   | Bayesian Information Criterion (BIC) . . . . .             | 78        |
| <b>12</b> | <b>Hypothesis Testing</b>                                  | <b>79</b> |
| 12.1      | Formulating Hypotheses . . . . .                           | 79        |
| 12.2      | Mechanics of a Hypothesis Test . . . . .                   | 80        |
| 12.2.1    | Test Statistic . . . . .                                   | 80        |
| 12.2.2    | Rejection Region . . . . .                                 | 80        |
| 12.2.3    | Significance Level and p-value . . . . .                   | 80        |
| 12.3      | Power of a Test . . . . .                                  | 80        |
| 12.4      | Wald Test . . . . .                                        | 81        |
| 12.4.1    | Hypothesis Setup . . . . .                                 | 81        |
| 12.5      | Definition of the Wald Test . . . . .                      | 81        |
| 12.5.1    | Alternative Version . . . . .                              | 81        |
| 12.5.2    | Type I Error Probability (Size) . . . . .                  | 81        |
| 12.5.3    | Power of the Wald Test . . . . .                           | 82        |
| 12.6      | Definition and Interpretation of p-Values . . . . .        | 82        |
| 12.7      | Multiple Testing . . . . .                                 | 82        |
| 12.8      | Mechanics of Multiple Testing . . . . .                    | 83        |
| 12.8.1    | Error Control Strategies . . . . .                         | 83        |
| 12.8.2    | False Discovery Rate (FDR) . . . . .                       | 83        |
| 12.9      | Likelihood Ratio Test (LRT) . . . . .                      | 84        |
| 12.9.1    | Asymptotic Approximation . . . . .                         | 84        |
| 12.10     | Goodness-of-Fit Tests . . . . .                            | 84        |
| 12.10.1   | Chi-Square Test . . . . .                                  | 84        |
| 12.10.2   | Kolmogorov-Smirnov Test . . . . .                          | 84        |

|                                             |    |
|---------------------------------------------|----|
| 12.11 Two-Sample Permutation Test . . . . . | 85 |
|---------------------------------------------|----|

# Chapter 1

## Introduction: 26/09/2024

### 1.1 Sample Space

The **sample space** is the set of all possible experimental outcomes. Usually denoted with the Greek letter  $\Omega$ .

$$\Omega = \{\text{Certain event}\}, \quad e, \quad \emptyset = \{\text{Impossible event}\}.$$

**Important:** Exploiting logical operations on sets (events), we can define new sets that correspond to events of interest. Let  $A, B$  be elements of  $\Omega$ :  $A, B \subseteq \Omega$ .

#### 1.1.1 Union of Sets

The event  $(A \cup B)$  is the subset of  $\Omega$  consisting of elements that belong to  $A$ , or  $B$ , or both; that is, it is the event that occurs when at least one between  $A$  and  $B$  occurs.

#### 1.1.2 Intersection of Sets

The event  $(A \cap B)$  is the subset of  $\Omega$  consisting of elements that belong to  $A$  and  $B$ ; that is, it is the event that occurs when both  $A$  and  $B$  occur simultaneously.

**Remark:** We say that two events are **disjoint** whenever they cannot occur simultaneously; that is, if the intersection of the corresponding sets is the empty set:  $(A \cap B) = \emptyset$ .

#### 1.1.3 Complement of a Set

$A^c$  denotes the subset of  $\Omega$  consisting of all the elements of  $\Omega$  not contained in  $A$ ; that is, the event that occurs when  $A$  does not occur.

#### 1.1.4 Venn Diagrams

Here are some basic set operations visualized using Venn diagrams.

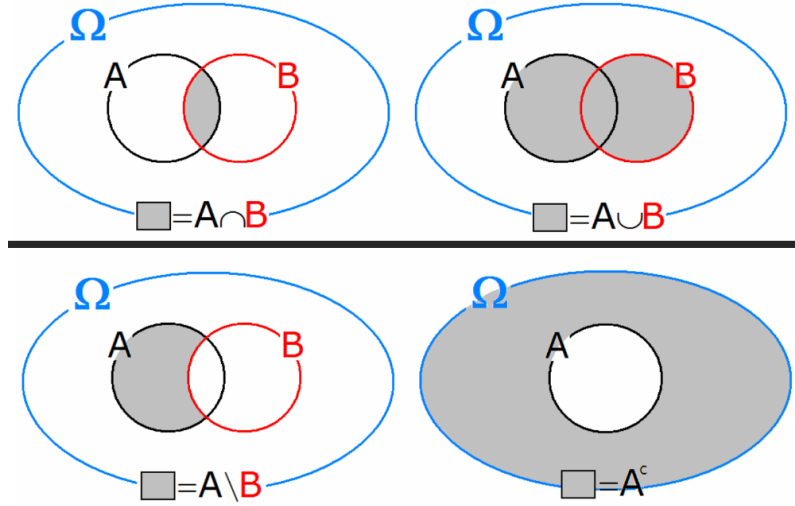


Figure 1.1: Venn diagrams representing set operations like  $A \cap B$ ,  $A \cup B$ ,  $A \setminus B$ , and  $A^c$  in the sample space  $\Omega$ .

**Remark:** For any  $A \subseteq \Omega$ , we have:

$$(A \cup A^c) = \Omega$$

$$(A \cup \Omega) = \Omega$$

$$(A \cup \emptyset) = A$$

$$(A \cap \Omega) = A$$

$$(A \cap \emptyset) = \emptyset$$

$A \subseteq B$  if and only if all elements of  $A$  belong also to  $B$ .

## 1.2 De Morgan's Laws

For any sequence of  $n$  events  $\{A_1, A_2, \dots, A_n\}$ :

$$(A_1 \cup A_2 \cup \dots \cup A_n)^c = A_1^c \cap A_2^c \cap \dots \cap A_n^c$$

$$(A_1 \cap A_2 \cap \dots \cap A_n)^c = A_1^c \cup A_2^c \cup \dots \cup A_n^c$$

We will see that it is extremely useful to express an event as a union of two disjoint events: Let  $A, B \subseteq \Omega$ ; then we have:

$$A \cap \Omega = A \cap (B \cup B^c) = (A \cap B) \cup (A \cap B^c).$$

## 1.3 Elementary Probability

### 1.3.1 Classical Definition

The probability of an event is the ratio of the number of cases favorable to it, to the number of all cases possible when nothing leads us to expect that any one of these cases should occur more than any other, which renders them equally likely.

$$P(E) = \frac{\text{Number of cases favorable to } E}{\text{Number of all possible cases}} = \frac{n_E}{n}$$



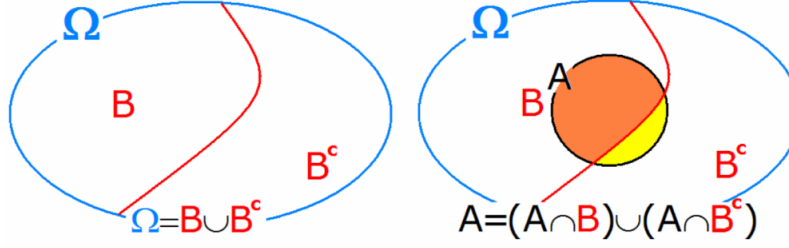


Figure 1.2: Venn diagrams illustrating  $\Omega = B \cup B^c$  and  $A = (A \cap B) \cup (A \cap B^c)$  in the sample space  $\Omega$ .

### 1.3.2 Frequentist Definition

Assume that an experiment can be repeated indefinitely under the same circumstances. Then, in the frequentist interpretation of probability, we define the probability of  $E$  as the limit of its relative frequency as the number of trials approaches infinity:

$$P(E) = \lim_{n \rightarrow \infty} \frac{n_E}{n}$$

*Frequentist Statistics and Scientific Replicability.*

### 1.3.3 Axiomatic Definition (Take 1)

A probability function or measure  $P(\cdot)$  on a finite sample space  $\Omega$  assigns to each event  $A \subseteq \Omega$  a number  $P(A)$  such that:

- $P(A) \in [0, 1]$ ;
- $P(\Omega) = 1$ ;
- $P(A \cup B) = P(A) + P(B)$  if  $A$  and  $B$  are disjoint events (finite additivity).

The number  $P(A)$  is called the probability that  $A$  occurs.

**Some consequences:**

$$P(A^c) = 1 - P(A)$$

## 1.4 Inclusion-Exclusion Principle

### 1.4.1 Two Events

$$P(A_1 \cup A_2) = P(A_1) + P(A_2) - P(A_1 \cap A_2).$$

### 1.4.2 Three Events

$$P(A_1 \cup A_2 \cup A_3) = P(A_1) + P(A_2) + P(A_3) - [P(A_1 \cap A_2) + P(A_2 \cap A_3) + P(A_3 \cap A_1)] + P(A_1 \cap A_2 \cap A_3).$$

**Proof: (Visual)**

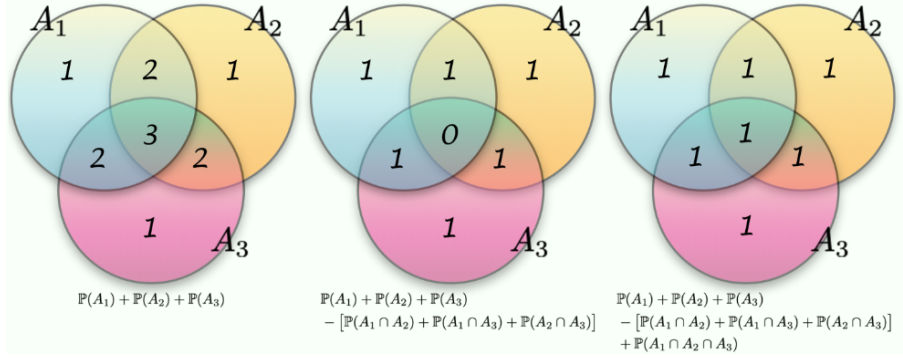


Figure 1.3: Visual proof of the Inclusion-Exclusion Principle with three events  $A_1$ ,  $A_2$ , and  $A_3$ .

## 1.5 Boole's Inequality (a.k.a. Union Bound)

For any finite or countable sequence of events  $\{A_1, A_2, \dots\}$ , we get that the probability of their union is bounded by the sum of individual probabilities:

$$P\left(\bigcup_i A_i\right) \leq \sum_i P(A_i)$$

**Proof (Induction):** For  $n = 1$ , it trivially follows that  $P(A_1) \leq P(A_1)$ . Now assume the result is true for a generic  $n$ ; that is,  $P(\bigcup_{i=1}^n A_i) \leq \sum_{i=1}^n P(A_i)$ , and let's prove it for  $n + 1$ . From the inclusion-exclusion principle (for 2 events) we get:

$$P\left(A_{n+1} \cup \bigcup_{i=1}^n A_i\right) = P(A_{n+1}) + P\left(\bigcup_{i=1}^n A_i\right) - P\left(A_{n+1} \cap \bigcup_{i=1}^n A_i\right)$$

Thus, by inductive hypothesis, we conclude:

$$P\left(\bigcup_{i=1}^{n+1} A_i\right) \leq \sum_{i=1}^{n+1} P(A_i)$$

## 1.6 Bonferroni's Inequality

For any sequence of events  $\{A_1, A_2, \dots, A_n\}$ , we get:

$$P\left(\bigcap_{i=1}^n A_i\right) \geq \sum_{i=1}^n P(A_i) - (n - 1)$$

**Proof** (using induction and inclusion-exclusion): Starting from:

$$P\left(\bigcap_{i=1}^n A_i\right) = 1 - P\left(\bigcup_{i=1}^n A_i^c\right)$$

By De Morgan's law and using Boole's inequality for the complement, we obtain the desired result.

## 1.7 Conditional Probability

We can define the conditional probability of  $(A|B)$  (the event that  $A$  occurs given that the event  $B$  is known to have occurred) as:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad \text{where } P(B) > 0.$$

Now that we know  $B \subseteq \Omega$  has occurred, only outcomes that are elements of  $B$  are possible. The conditional probability is the probability of falling within the intersection of  $A$  and  $B$ , normalized by dividing by  $P(B)$ .

### Conditional Probability Visualization



Figure 1.4: Venn diagram representing  $A \cap B$  under different visualizations.

For any  $A \subseteq \Omega$ , the function  $A \mapsto P(A|B)$  is a new legit probability measure that obeys the usual axioms and rules:

1.  $P(A|B) \geq 0$ ;
2.  $P(\Omega|B) = 1$ ;
3.  $P\left(\bigcup_{i=1}^n A_i \middle| B\right) = \sum_{i=1}^n P(A_i|B)$  for any sequence of disjoint events  $\{A_1, A_2, \dots\}$ .

## 1.8 Multiplication Rule

The joint probability of two events  $A$  and  $B$  to occur is given by:

$$P(A \cap B) = P(A|B) \cdot P(B) = P(B|A) \cdot P(A).$$

## 1.9 Chain Rule

Given a sequence of events  $\{A_1, A_2, \dots, A_n\}$ , the repeated application of the multiplication rule allows one to calculate the joint probability:

$$P(A_1 \cap A_2 \cap \dots \cap A_n) = P(A_1) \cdot P(A_2|A_1) \cdot P(A_3|A_1 \cap A_2) \dots$$

## 1.10 Independence

Two events  $A$  and  $B$  are independent, written as  $A \perp B$ , if and only if:

$$P(A \cap B) = P(A) \cdot P(B).$$

We say that events  $A$  and  $B$  are conditionally independent given an event  $C$ , written as  $A \perp B|C$ , if and only if:

$$P(A \cap B|C) = P(A|C) \cdot P(B|C).$$

**Pairwise Independence:** Events  $\{A_i\}_{i \in \mathbb{N}}$  are pairwise independent if for any  $i \neq j$ , we have:

$$P(A_i \cap A_j) = P(A_i) \cdot P(A_j).$$

**Mutual Independence:** Events  $\{A_i\}_{i \in \mathbb{N}}$  are mutually independent if for any subset of events  $A_{i_1}, A_{i_2}, \dots, A_{i_k}$ , we have:

$$P(A_{i_1} \cap A_{i_2} \cap \dots \cap A_{i_k}) = P(A_{i_1}) \cdot P(A_{i_2}) \cdot \dots \cdot P(A_{i_k}).$$

## 1.11 The Law of Total Probabilities

For any pair of events  $A$  and  $B$  in  $\Omega$ , we have:

$$P(A) = P(A|B) \cdot P(B) + P(A|B^c) \cdot P(B^c).$$

**Proof:** Consider the usual decomposition of  $\Omega$  induced by  $B$ :

$$\Omega = B \cup B^c.$$

Hence, using the union of two disjoint events:

$$A = (A \cap \Omega) = (A \cap (B \cup B^c)) = (A \cap B) \cup (A \cap B^c).$$

Therefore,

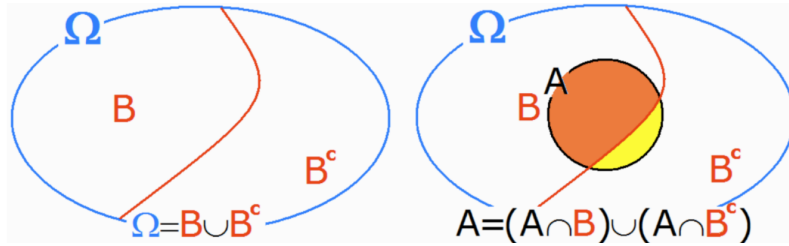


Figure 1.5: Venn diagram illustrating  $\Omega = B \cup B^c$  and  $A = (A \cap B) \cup (A \cap B^c)$  in the sample space  $\Omega$ .

$$P(A) = P((A \cap B) \cup (A \cap B^c)) = P(A \cap B) + P(A \cap B^c) = P(A|B) \cdot P(B) + P(A|B^c) \cdot P(B^c).$$

For any given partition  $\mathcal{A} = \{A_1, \dots, A_n\}$  of  $\Omega$  and a specific event  $E$  of interest, we have:

$$P(E) = \sum_{i=1}^n P(E|A_i) \cdot P(A_i).$$

## 1.12 Bayes' Theorem

For a given event  $E$  and a set of hypotheses  $H_i$  in  $\Omega$ , we can express the conditional probability of a hypothesis given the evidence:

$$P(H_i|E) = \frac{P(E|H_i) \cdot P(H_i)}{P(E)}.$$

Where:

- $P(E|H_i)$  is the **Likelihood**: How probable is the evidence given that our hypothesis is true?
- $P(H_i)$  is the **Prior**: How probable was our hypothesis before observing the evidence?
- $P(E)$  is the **Marginal**: How probable is the evidence under all possible hypotheses?

Posterior Odds can also be defined as:

$$O(H_i|E) = \frac{P(H_i|E)}{P(H_i^c|E)} = \text{BF} \times O(H_i),$$

where BF is the Bayes Factor.

The main idea of Bayesian learning is to use probability to represent our state of uncertainty about unknown phenomena. We update our beliefs based on observable data using the rules of probability. Prior beliefs about an unknown parameter are represented as a **prior distribution**  $\pi(\theta)$ . Once data is observed, we update to the **posterior distribution** using Bayes' rule:

$$\pi(\theta|x) = \frac{f(x|\theta)\pi(\theta)}{m(x)}.$$

### 1.12.1 Probability and Statistics

The main assumption underlying statistical inference is that there are unknown generating mechanisms behind the observed data. More specifically, we assume probabilistic mechanisms govern the results of our experiments.

**Statistical Inference:** Formalizes the inductive process that allows learning unknown features of a population based on sample observations. The sample represents a subset of the population, and based on the results, we estimate the probabilistic laws generating the population.

A **Statistical Experiment** is a process where:

- Before the experiment, the outcome is unknown.
- After the experiment, we know that one of the possible outcomes occurred.

# Chapter 2

## Random Variables:01/10/2024(4a)

### 2.1 Remark

#### Distribution of $X$

The probability that  $X$  belongs to a set  $E \subseteq \mathbb{R}$  is:

$$P(X \in E) = P(\{\omega \in \Omega : X(\omega) \in E\}),$$

where  $\omega$  represents the outcomes in the sample space  $\Omega$ .

#### Inverse

For any set  $E \subseteq \mathbb{R}$ , the inverse image  $X^{-1}(E)$  is defined as:

$$X^{-1}(E) = \{\omega \in \Omega : X(\omega) \in E\}.$$

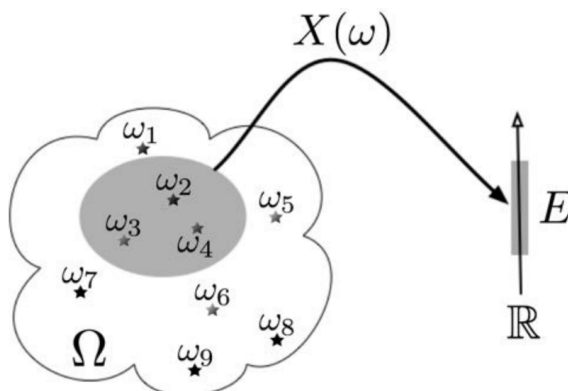
This set represents all the outcomes  $\omega$  that map to values within  $E$  under  $X$ .

### 2.2 Random Variables and Distributions

**Random Variable:** Given a probability space  $(\Omega, \mathcal{F}, P)$ , a random variable is any measurable mapping:

$$X : \Omega \rightarrow \mathbb{R}.$$

The random variable assigns a real number to each outcome  $\omega \in \Omega$ .



The distribution of a random variable can be described using:

- **Probability Mass Function (PMF):** For discrete random variables, it defines probabilities for discrete outcomes:

$$p_X(x) = P(X = x).$$

- **Probability Density Function (PDF):** For continuous variables:

$$\mathbb{P}(a \leq X \leq b) = \int_a^b f_X(x) dx.$$

**N.B.** using the PMF for continuous R.var gives values close to 0

- The **Cumulative Distribution Function (CDF)** for a random variable  $X$  is the function  $F_X(x) : \mathbb{R} \rightarrow [0, 1]$  defined by

$$F_X(x) = \mathbb{P}(X \leq x) = \mathbb{P}(X \in (-\infty, x]) \quad \forall x \in \mathbb{R}.$$

## 2.3 Basic Properties

### 2.3.1 Basic Properties of PMF

1. **Non-Negativity:**

$$p_X(x) \geq 0 \quad \text{for all } x \in \mathbb{R}.$$

The probability of any specific value must be non-negative.

2. **Normalization:**

$$\sum_{x \in \text{Support of } X} p_X(x) = 1.$$

The sum of the probabilities over all possible values of the discrete random variable must equal 1. This represents the fact that one of the possible outcomes must occur.

3. **Probability of Individual Events:**

$$p_X(x) = P(X = x).$$

The PMF assigns the probability of each possible value that  $X$  can take.

4. **Support of  $X$ :** The support of a discrete random variable  $X$  is the set of all values for which  $p_X(x) > 0$ .

### 2.3.2 Basic Properties of PDF

1. **Non-Negativity:**

$$f_X(x) \geq 0 \quad \text{for all } x \in \mathbb{R}.$$

2. **Support of  $X$ :** The support of  $X$ — $\text{supp}(X)$ —is the set of values  $x \in \mathcal{X}$  such that  $f_X(x) > 0$ .

3. **Normalization:**

$$\int_{\mathbb{R}} f_X(x) dx = 1.$$

### 2.3.3 Basic Properties of CDF

- The CDF  $F_X(x)$  of a random variable  $X$  represents the probability that  $X$  takes a value less than or equal to  $x$ :

$$F_X(x) = P(X \leq x)$$

- The CDF  $F_X(x)$  is a **non-decreasing function**, meaning that for any  $x_1 < x_2$ :

$$F_X(x_1) \leq F_X(x_2)$$

- The CDF is **normalized**:

$$\lim_{x \rightarrow -\infty} F_X(x) = 0, \quad \lim_{x \rightarrow +\infty} F_X(x) = 1$$

- The CDF is **right-continuous**, meaning that for all  $x$ :

$$F_X(x) = F_X(x^+) = \lim_{t \rightarrow x^+} F_X(t)$$

- The CDF is bounded between 0 and 1:

$$0 \leq F_X(x) \leq 1 \quad \forall x \in \mathbb{R}$$

#### Additional Properties (Discrete vs Continuous)

- **For discrete random variables:** The CDF  $F_X(x)$  is a step function, and the probability for  $X = x$  can be calculated as the difference between the CDF values:

$$P(X = x) = F_X(x) - F_X(x^-)$$

- **For continuous random variables:** The CDF is differentiable, and the PDF  $f_X(x)$  is the derivative of the CDF:

$$f_X(x) = \frac{d}{dx} F_X(x)$$

Additionally:

$$F_X(x) = \int_{-\infty}^x f_X(t) dt$$



## 2.4 Example: Fair Dice (Discrete Case)

Consider a fair dice with  $P(X = x) = \frac{1}{6}$  for  $x \in \{1, 2, 3, 4, 5, 6\}$ .

For the **PMF**, the graph is a flat line where each point (1 through 6) has a probability of  $\frac{1}{6}$ , represented by a dot on the y-axis. This is a typical flat-line PMF.

For the **CDF**, we begin at  $F_X(x) = 0$  from  $-\infty$  until the first event (e.g.,  $X = 1$ ) occurs. Then, we add the probabilities cumulatively:

$$F_X(x) = \begin{cases} 0 & x < 1, \\ \frac{1}{6} & 1 \leq x < 2, \\ \frac{2}{6} & 2 \leq x < 3, \\ \frac{3}{6} & 3 \leq x < 4, \\ \frac{4}{6} & 4 \leq x < 5, \\ \frac{5}{6} & 5 \leq x < 6, \\ 1 & x \geq 6. \end{cases}$$

This produces a step-function graph that increases as we include each event until it reaches 1, and remains constant for all  $x \geq 6$ .

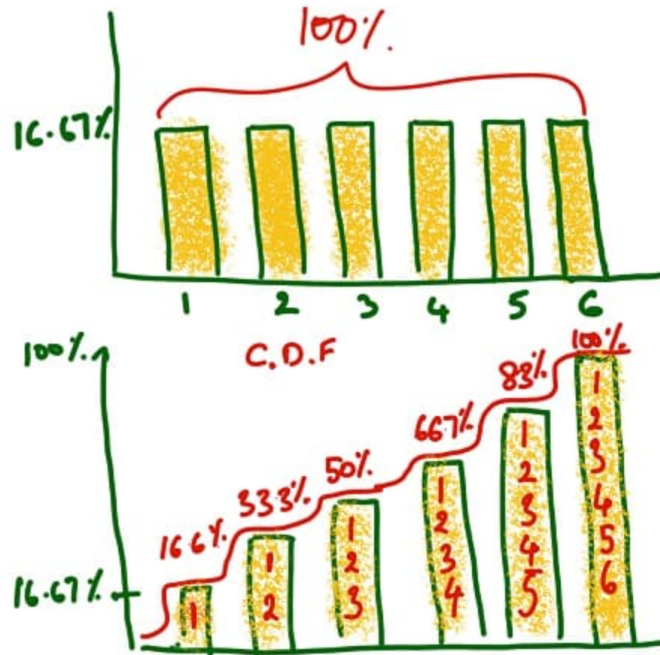


Figure 2.1: Visual representation of PMF (top) and CDF (bottom) for a fair dice. The PMF shows equal probability for each face of the dice, while the CDF accumulates the probability as each face is considered.

### Moving from PMF to CDF

To move from the PMF to the CDF for a discrete random variable, we simply sum the probabilities up to the point  $x_i$  of interest:

$$F_X(x) = \sum_{x_i \leq x} P(X = x_i).$$

In a continuous random variable, this would instead involve integrating the probability density function (PDF):

$$F_X(x) = \int_{-\infty}^x f_X(t) dt.$$

## Modeling Perspective: Continuous Random Variables and Intervals

In the case of continuous random variables, it is more meaningful to consider intervals rather than single points. The probability of observing a specific value  $X = x$  is always zero, but we can compute the probability that  $X$  lies within a small interval, e.g.,  $[x, x + \delta x]$ , where  $\delta x$  is a small value approaching 0.

The key points in this context are:

- **Right Continuity of the CDF:** The CDF  $F_X(x)$  is right-continuous and non-decreasing.
- **Shrinking Intervals:** The probability of  $X$  lying within an interval  $[x, x + \delta x]$  is approximated by:

$$P(x \leq X \leq x + \delta x) \approx f_X(x) \cdot \delta x.$$

As  $\delta x \rightarrow 0$ , this converges to the definition of the PDF.

## Transition from CDF to PMF

To transition back from the CDF to the PMF (for discrete random variables), we look at the jumps in the CDF. Each jump corresponds to the probability of a particular outcome. The size of the jump at  $x_i$  equals  $P(X = x_i)$ .

## 2.5 The Quantile Function

Let  $X$  be any random variable (continuous or discrete) with CDF  $F_X(\cdot)$ . Then, for any given level (“tail area”)  $p \in [0, 1]$ , the quantile function is:

$Q(p) = \inf\{F_X(x) \geq p\}$  = The smallest value  $x$  such that  $F_X(x) = P(X \leq x)$  is at least  $p$ .

As a final remark, let’s explicitly notice that for a **continuous random variable**  $X$  with CDF  $F_X(\cdot)$  and PDF  $f_X(\cdot)$ , the quantile at level  $p$  is the value  $q_p$  that solves the following equation:

$$p = F_X(q_p) = \int_{-\infty}^{q_p} f_X(t) dt.$$

- **Median:**  $q_{0.50}$  is the smallest number such that:

$$F_X(q_{0.50}) \geq 0.50, \quad q_{0.50} = 0$$

- **90th percentile:**  $q_{0.90}$  is the smallest number such that:

$$F_X(q_{0.90}) \geq 0.90, \quad q_{0.90} = 1$$

- **3rd quartile:**  $q_{0.75}$  is the smallest number such that:

$$F_X(q_{0.75}) \geq 0.75, \quad q_{0.75} = 1$$

# Chapter 3

## Univariate Distributions(4b)

### 3.1 Bernoulli Distribution

A discrete random variable  $X$  has a Bernoulli distribution with parameter  $p \in [0, 1]$ , if its probability mass function (pmf) is:

$$p_X(1) = P(X = 1) = p \quad \text{and} \quad p_X(0) = P(X = 0) = 1 - p$$

In one line, we can write it as:

$$p_X(x) = p^x(1 - p)^{1-x}, \quad x \in \{0, 1\}$$

We denote this by  $X \sim \text{Ber}(p)$ .

#### Applications

The Bernoulli distribution models binary experiments, where there are only two possible outcomes, typically labeled as “success” and “failure.” The parameter  $p$  is the probability of success. A typical example is flipping a coin, where success could be getting “heads.”

### 3.2 Binomial Distribution

A discrete R. var  $X$  follows a Binomial distribution with parameters  $n \in \mathbb{N}_0$  (number of trials) and  $p \in [0, 1]$  (probability of success in each trial), if its pmf is:

$$p_X(x) = P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}, \quad x \in \{0, 1, \dots, n\}$$

We denote  $X \sim \text{Binom}(n, p)$ .

#### Applications

The Binomial distribution models the number of successes in  $n$  independent Bernoulli trials. The distribution can be used in many real-world scenarios where there are repeated independent trials with the same probability of success.

### PMF & CDF

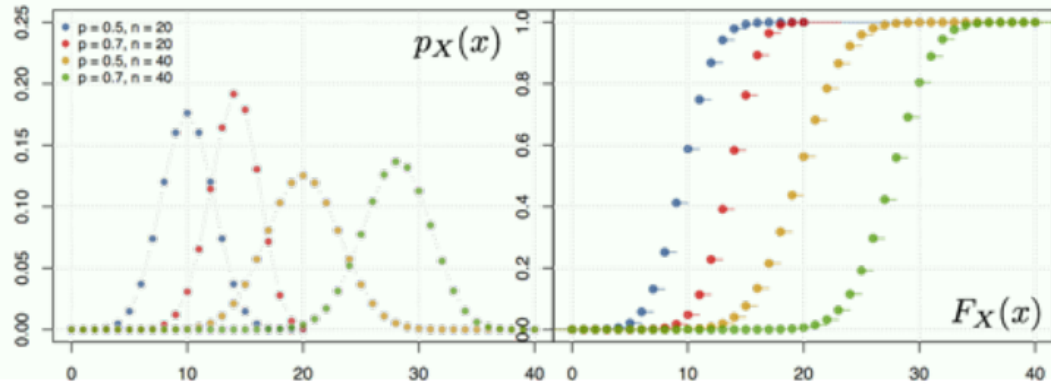


Figure 3.1: Example of Binomial Distribution

## 3.3 Geometric Distribution

A discrete random variable  $X$  follows a Geometric distribution with parameter  $p \in [0, 1]$ , if its pmf is:

$$p_X(x) = P(X = x) = p(1 - p)^{x-1}, \quad x \in \{1, 2, 3, \dots\}$$

We denote  $X \sim \text{Geom}(p)$ .

### Applications

The Geometric distribution models the number of trials needed to get the first success in repeated Bernoulli trials. For instance, the number of times you need to flip a coin before you get the first “heads.” This distribution is memoryless, meaning the probability of success in the future is independent of past outcomes.

### PMF & CDF

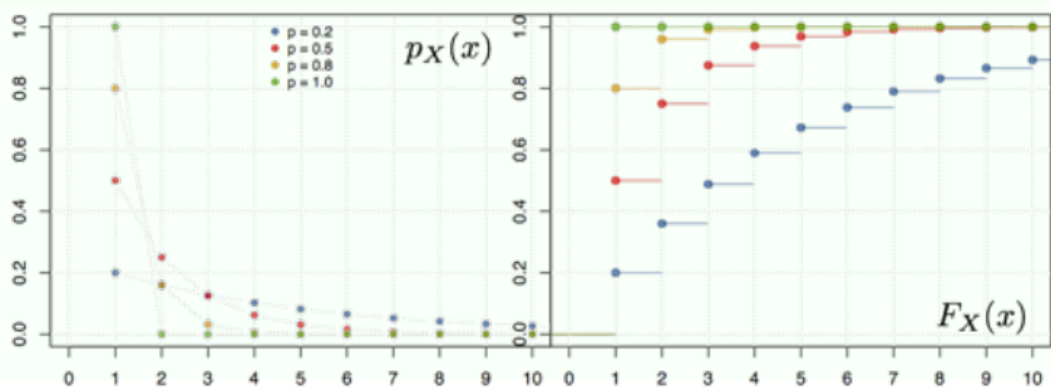


Figure 3.2: Example of Geometric distribution

### 3.4 Poisson Distribution

A discrete random variable  $X$  follows a Poisson distribution with parameter  $\mu > 0$  (average number of events per interval), if its pmf is:

$$p_X(x) = P(X = x) = \frac{\mu^x e^{-\mu}}{x!}, \quad x \in \{0, 1, 2, \dots\}$$

We denote  $X \sim \text{Pois}(\mu)$ .

#### Applications

The Poisson distribution models the number of rare events occurring in a fixed interval of time or space. Examples include the number of emails received in an hour or the number of cars passing through a toll booth in a given period.

#### Example

If you expect to receive an average of 5 emails per hour, then  $X \sim \text{Pois}(\mu = 5)$ . The probability of receiving exactly 3 emails in one hour is:

$$P(X = 3) = \frac{5^3 e^{-5}}{3!}$$

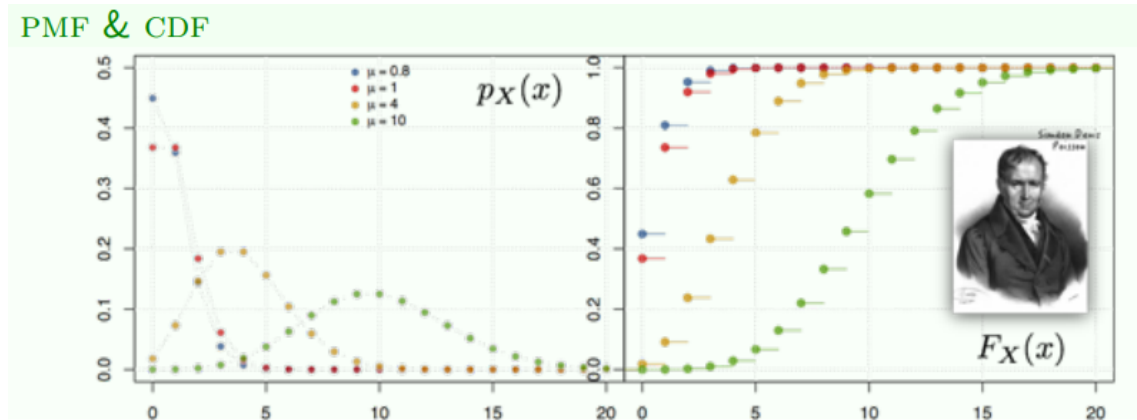


Figure 3.3: Example of Poisson Distribution

### 3.5 Uniform Distribution

A continuous random variable  $X$  follows a Uniform distribution on the interval  $[a, b]$  if its probability density function (pdf) is:

$$f_X(x) = \frac{1}{b - a} \quad \text{for } x \in [a, b]$$

We denote  $X \sim \text{Unif}(a, b)$ .

## Applications

The Uniform distribution assigns equal probability to every point within the interval  $[a, b]$ . It is often used in simulations when generating random numbers

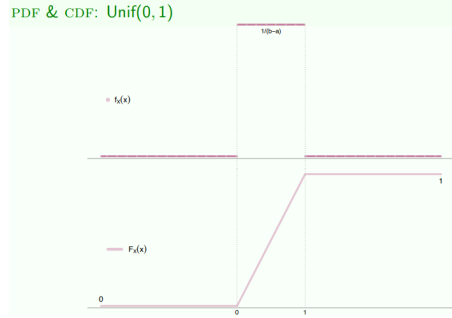


Figure 3.4: Example of Distribution of Uniform

## 3.6 Exponential Distribution

A continuous random variable  $X$  follows an Exponential distribution with parameter  $\lambda > 0$  (rate of occurrence), if its pdf is:

$$f_X(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

We denote  $X \sim \text{Exp}(\lambda)$ .

## Applications

The Exponential distribution is often used to model waiting times, such as the time until the next customer arrives in a queue. It is memoryless, meaning the future waiting time is independent of the time already waited.

## Example

If the average time between customer arrivals is 5 minutes, then  $\lambda = \frac{1}{5}$ . The probability that the next customer will arrive within 3 minutes is:

$$P(X \leq 3) = 1 - e^{-\lambda \cdot 3} = 1 - e^{-\frac{3}{5}}$$

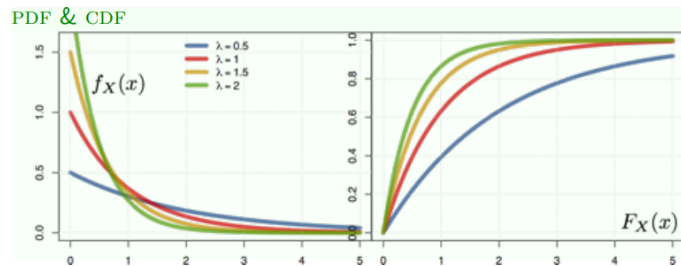


Figure 3.5: Example of Exponential distribution

### 3.7 Normal/Gaussian Distribution

A continuous random variable  $X$  follows a Normal (or Gaussian) distribution with mean  $\mu$  and variance  $\sigma^2$ , if its pdf is:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R}$$

We denote  $X \sim N(\mu, \sigma^2)$ .

Standard Normal Distribution has  $(\mu, \sigma^2) = (0, 1)$ ,  $Z = \frac{X-\mu}{\sigma} \sim N(0, 1)$

#### From $N(0,1)$ to $N(\mu, \sigma^2)$

We can compute **any** probabilities we want under a generic  $N(\mu, \sigma^2)$  as long as we can compute the **CDF** of a standard Normal. In fact, for any  $a < b \in \mathbb{R}$ , we have:

$$\begin{aligned} \mathbb{P}(X \in (a, b)) &= \mathbb{P}(a < X < b) = \mathbb{P}\left(\frac{a-\mu}{\sigma} < \frac{X-\mu}{\sigma} < \frac{b-\mu}{\sigma}\right) \\ &= \mathbb{P}\left(\frac{a-\mu}{\sigma} < Z < \frac{b-\mu}{\sigma}\right) \\ &= \Phi\left(\frac{b-\mu}{\sigma}\right) - \Phi\left(\frac{a-\mu}{\sigma}\right) \end{aligned}$$

where  $\Phi$  is the CDF of the standard Normal distribution.

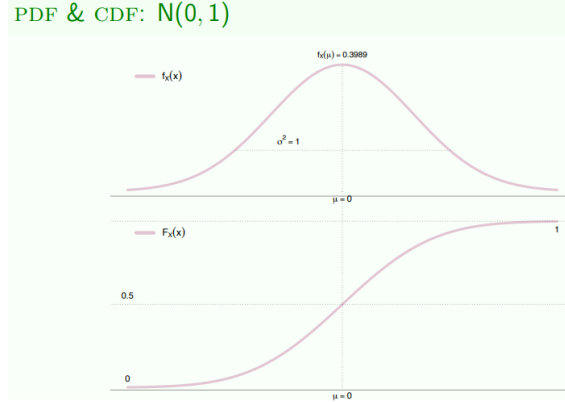


Figure 3.6: Example of Normal distribution

### 3.8 Student's t-Distribution

A continuous random variable  $X$  follows a Student's t-distribution with  $\nu$  degrees of freedom, if its pdf is:

$$f_X(x) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi}\Gamma\left(\frac{\nu}{2}\right)} \left(1 + \frac{x^2}{\nu}\right)^{-\frac{\nu+1}{2}}, \quad x \in \mathbb{R}$$

We denote  $X \sim t(\nu)$ .

## Properties

- The distribution is symmetric with heavier tails compared to the Normal distribution.

- As  $\nu$  increases, the t-distribution approaches the Normal distribution.
- For  $\nu = 1$ , it is known as the Cauchy distribution.

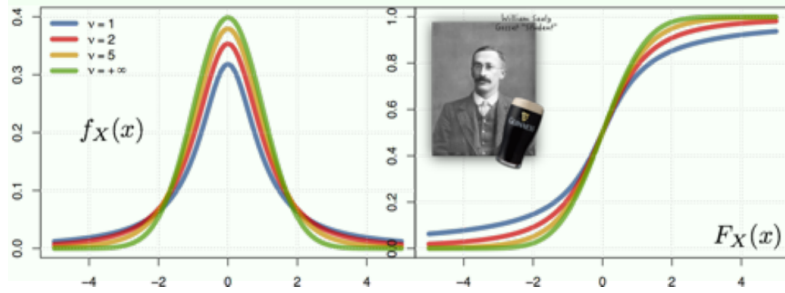


Figure 3.7: Example of Student's t-distribution

## 3.9 Laplace Distribution

A continuous random variable  $X$  has a Laplace distribution with parameters  $\mu$  (location) and  $\lambda > 0$  (scale), if its pdf is:

$$f_X(x) = \frac{1}{2\lambda} \exp\left(-\frac{|x - \mu|}{\lambda}\right), \quad x \in \mathbb{R}$$

We denote  $X \sim \text{Laplace}(\mu, \lambda)$ .

## Properties

- The Laplace distribution has heavier tails than the Normal distribution.
- It is often used in modeling data with a sharp peak at the mean.

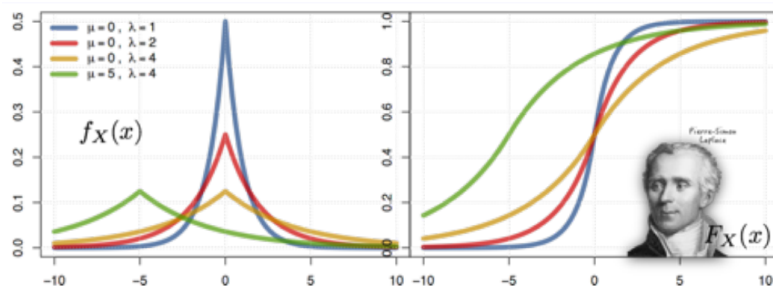
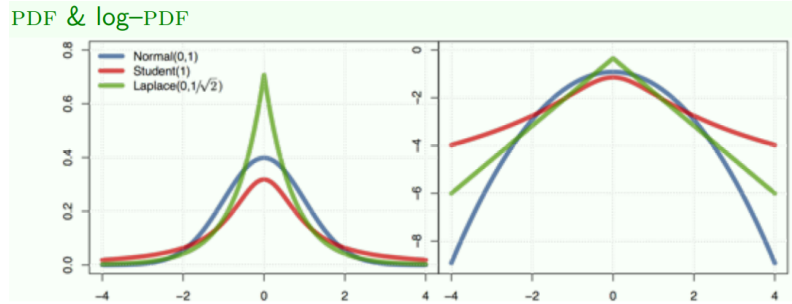


Figure 3.8: Example of Laplace distribution



### 3.10 Comparison between Normal, T-student and Laplacian



- **Tails and Outliers:** The Student's t and Laplace distributions have heavier tails than the Normal, making them better for modeling outliers.
- **Shape of the Peak:** The Laplace distribution has a sharper peak at the mean, useful for capturing sudden changes.
- **Convergence to Normal:** The Student's t-distribution approaches the Normal distribution as degrees of freedom increase, providing flexibility.

### 3.11 Gamma Distribution

A continuous random variable  $X$  follows a Gamma distribution with shape parameter  $\alpha > 0$  and rate parameter  $\lambda > 0$ , if its pdf is:

$$f_X(x) = \frac{\lambda^\alpha x^{\alpha-1} e^{-\lambda x}}{\Gamma(\alpha)}, \quad x \geq 0$$

We denote  $X \sim \text{Gamma}(\alpha, \lambda)$ .

#### Special Cases

- When  $\alpha = 1$ , the Gamma distribution reduces to the Exponential distribution.
- The Gamma distribution is also related to the Chi-square distribution.

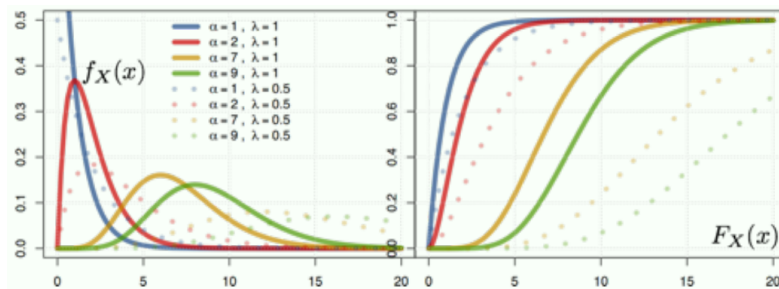


Figure 3.9: Example of Gamma distribution

## 3.12 Beta Distribution

A continuous random variable  $X$  has a Beta distribution with parameters  $\alpha, \beta > 0$ , if its pdf is:

$$f_X(x) = \frac{1}{B(\alpha, \beta)} x^{\alpha-1} (1-x)^{\beta-1}, \quad x \in [0, 1]$$

We denote  $X \sim \text{Beta}(\alpha, \beta)$ .

### Properties

- The Beta distribution is flexible for modeling random variables in the range  $[0, 1]$ .
- It is often used in Bayesian statistics as a prior distribution.

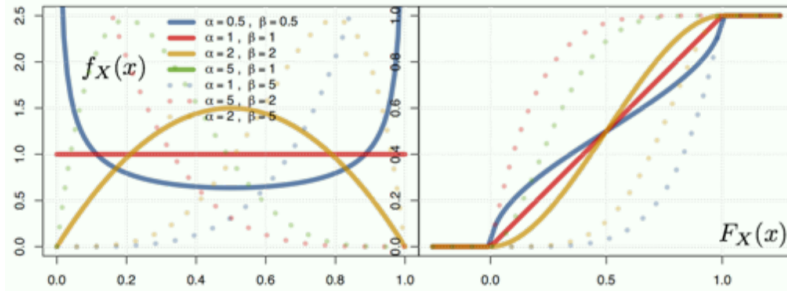


Figure 3.10: Example of Beta distribution

## 3.13 Exponential Family

The exponential family is a set of probability distributions whose PDF or PMF can be expressed in the form:

$$f_X(x | \theta) = h(x) \exp(\eta(\theta) \cdot T(x) - A(\theta))$$

where  $T(x)$ ,  $h(x)$ ,  $\eta(\theta)$ , and  $A(\theta)$  are known functions, and  $\theta$  is the parameter of the family.

### Examples

- Normal, Exponential, Gamma, Beta, Bernoulli, and Poisson distributions belong to the exponential family.
- The Student's t-distribution and Uniform distribution do not belong to the exponential family.

# Chapter 4

## Fam.Prob - Models: 04/10/2024

### 4.0.1 Learning Tasks and Loss Functions

For any learning task (e.g., classification, regression), we define a **loss function**. The choice of the loss function depends on the specific task, and sometimes more than one can be used. The **Data Generating Process (DGP)**, which includes the **Population Model** and the **Sampling Scheme**, provides enough information to choose the right loss function.

### 4.1 Logistic Regression Example

Binary classification: We aim to predict the binary response variable  $Y \in \{0, 1\}$  based on a set of covariates (features)  $X = [X_1, \dots, X_p] \in \mathbb{R}^p$ .

The joint distribution of  $(Y, X)$  is Case 1: All Discr. Rand. Var. modeled as:

$$F_{Y,X} = F_{Y|X} F_X$$

where  $F_{Y|X}$  is the conditional distribution of  $Y$  given  $X$ , modeled as a Bernoulli distribution:

$$Y \sim \text{Bern}(p(X))$$

where  $p(X)$  depends on the covariates.

#### 4.1.1 Modeling Logistic Regression

The logistic regression model can be written as:

$$p(X) = P(Y = 1|X) = \alpha + \langle \beta, X \rangle$$

where  $\alpha \in \mathbb{R}$  is the intercept and  $\beta \in \mathbb{R}^p$  are the coefficients of the covariates.

**Problem:** The right-hand side takes values in  $\mathbb{R}$ , but we need  $p(X)$  to be in the range  $[0, 1]$ .

To preserve linearity, we apply a **logistic function (sigmoid)** to map the linear predictor to a probability:

$$F(z) = \frac{1}{1 + e^{-z}}$$

where  $z = \alpha + \langle \beta, X \rangle$ .

### 4.1.2 Link Function and Logistic Function

The logistic function is a common **link function** used in binary classification models. For large negative  $z$ ,  $F(z) \approx 0$ , and for large positive  $z$ ,  $F(z) \approx 1$ . The **log-odds (logit function)** is:

$$\log \left( \frac{p(X)}{1 - p(X)} \right) = \alpha + \langle \beta, X \rangle$$

## 4.2 Sampling and Likelihood

The sampling scheme defines how data is collected, with the assumption that the data points are **independent and identically distributed (i.i.d.)**.

The **likelihood function** for Bernoulli-distributed data is:

$$P(Y_i = y_i) = p^{y_i}(1 - p)^{1 - y_i}, \quad i \in \{1, \dots, n\}$$

Given the independence assumption, the joint probability of observing the data is the product of individual probabilities:

$$L(p) = \prod_{i=1}^n p^{y_i}(1 - p)^{1 - y_i}$$

### 4.2.1 Maximizing the Likelihood

We maximize the **log-likelihood function**:

$$\log L = \sum_{i=1}^n (y_i \log p(X_i) + (1 - y_i) \log(1 - p(X_i)))$$

The best parameters  $\alpha$  and  $\beta$  are those that maximize the log-likelihood. This process is called **maximum likelihood estimation (MLE)**.

The optimal  $\alpha$  and  $\beta$  are found using optimization methods like **Gradient Descent (GD)** or **Stochastic Gradient Descent (SGD)**.

## 4.3 Random Vectors

A **random vector**  $X = [X_1, \dots, X_p] \in \mathbb{R}^p$  maps outcomes  $\omega \in \Omega$  to  $\mathbb{R}^p$ . The components  $X_1, \dots, X_p$  can be continuous or discrete.

### 4.3.1 Case 1: All Discrete Random Variables

Each component has a **support** (set of possible values), denoted  $\gamma(X_1), \gamma(X_2), \dots$ . The **joint probability mass function (PMF)** describes the probability of each combination of values from the supports:

$$P_{X_1, X_2}(x_1, x_2) = P[(X_1 = x_1, X_2 = x_2)]$$

For example: -  $X_1$  = Pass (Yes/No),  $X_2$  = Attend Class (Never/Sometimes/Often)

Joint probabilities are organized in a table:

| Pass/Attend | Never | Sometimes | Often | Sum |
|-------------|-------|-----------|-------|-----|
| Y           | 0.1   | 0.1       | 0.3   | 0.5 |
| N           | 0.4   | 0.1       | 0     | 0.5 |
| Sum         | 0.5   | 0.2       | 0.3   | 1   |

### Remarks

- **Marginals:** The marginal probability of  $X_1 = x_1$  is obtained by summing over the joint probabilities with respect to  $X_2$ :

$$P(X_1 = x_1) = \sum_{j=1}^m P(X_1 = x_{1j}, X_2 = x_{2j})$$

- **Conditionals:** The conditional probability describes how likely one event is given that another has occurred:

$$P(X_1 = x_{11} | X_2 = x_{21}) = \frac{P(X_1 = x_{11}, X_2 = x_{21})}{P(X_2 = x_{21})}$$

For example, the probability of passing given that the student never attends is:

$$P(\text{Pass} \mid \text{Never}) = \frac{0.1}{0.5} = 0.2$$

- **Independence:** Random variables  $X_1$  and  $X_2$  are independent if their joint probability equals the product of their marginals:

$$P(X_1 = x_1, X_2 = x_2) = P(X_1 = x_1)P(X_2 = x_2)$$

### 4.3.2 Joint PMF: Topic 09/10

For a bivariate discrete random vector  $\vec{X} = [X_1, X_2]$ , we define the joint probability mass function (PMF) as:

$$P_{X_1, X_2}(x_1, x_2) = P((X_1 = x_1) \cap (X_2 = x_2))$$

The sum of all probabilities in the joint PMF must equal 1:

$$\sum_{x_1} \sum_{x_2} P_{X_1, X_2}(x_1, x_2) = 1$$

### Marginalization

To derive the marginal distribution of  $X_2$ , we sum over all values of  $X_1$ :

$$P_{X_2}(x_2) = \sum_{x_1} P_{X_1, X_2}(x_1, x_2)$$

Similarly, for  $X_1$ :

$$P_{X_1}(x_1) = \sum_{x_2} P_{X_1, X_2}(x_1, x_2)$$

## Conditional Probability

The conditional probability of  $X_1$  given  $X_2$  is:

$$P_{X_1|X_2}(x_1|x_2) = \frac{P_{X_1,X_2}(x_1, x_2)}{P_{X_2}(x_2)}$$

### 4.3.3 Bivariate CDF

The joint cumulative distribution function (CDF) of a bivariate random vector  $\vec{X} = [X_1, X_2]$  is defined as:

$$F_{X_1,X_2}(x_1, x_2) = P((X_1 \leq x_1) \cap (X_2 \leq x_2))$$

This can also be written as a double sum over the joint PMF:

$$F_{X_1,X_2}(x_1, x_2) = \sum_{s \leq x_1} \sum_{t \leq x_2} P_{X_1,X_2}(s, t)$$

### Marginal CDF

Taking limits of the joint CDF gives the marginal CDFs:

$$\lim_{x_1 \rightarrow +\infty} F_{X_1,X_2}(x_1, x_2) = F_{X_2}(x_2)$$

$$\lim_{x_2 \rightarrow +\infty} F_{X_1,X_2}(x_1, x_2) = F_{X_1}(x_1)$$

The limits at infinity give us:

$$\lim_{x_1, x_2 \rightarrow \infty} F_{X_1,X_2}(x_1, x_2) = 1$$

$$\lim_{x_1, x_2 \rightarrow -\infty} F_{X_1,X_2}(x_1, x_2) = 0$$

### 4.3.4 Joint PDF (Continuous Case)

In the continuous case, for a bivariate random vector  $\vec{X} = [X_1, X_2]$ , the joint probability density function (PDF)  $f_{X_1,X_2}(x_1, x_2)$  satisfies:

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X_1,X_2}(x_1, x_2) dx_1 dx_2 = 1$$

### Marginalization (Continuous Case)

To obtain the marginal distribution for  $X_1$ :

$$f_{X_1}(x_1) = \int_{-\infty}^{\infty} f_{X_1,X_2}(x_1, x_2) dx_2$$

### Conditional PDF (Continuous Case)

The conditional PDF of  $X_1$  given  $X_2$  is defined as:

$$f_{X_1|X_2}(x_1|x_2) = \frac{f_{X_1,X_2}(x_1, x_2)}{f_{X_2}(x_2)}$$

#### 4.3.5 Example

$$F_{X,Y}(x, y) = s - e^{-\alpha x} - e^{-\beta y} + e^{-(\alpha x + \beta y)}$$

where  $x, y > 0$  and  $\alpha, \beta > 0$ .

### Conditional Distribution

**Option 1:** Joint CDF  $\rightarrow$  Joint PDF  $\rightarrow$  Conditional

$$\begin{aligned} f_{X,Y}(x, y) &= \frac{\partial^2}{\partial x \partial y} F_{X,Y}(x, y) = \frac{\partial}{\partial y} [\alpha x e^{-\alpha x} - \alpha x e^{-(\alpha x + \beta y)}] \\ &= \alpha \beta e^{-(\alpha x + \beta y)} \cdot \mathbb{I}_{\mathbb{R}_+^2}(x, y) \end{aligned}$$

$$f_X(x) = \int_{\mathbb{R}} f_{X,Y}(x, y) dy = \int_0^{+\infty} \alpha \beta e^{-(\alpha x + \beta y)} dy = \alpha e^{-\alpha x} \cdot \mathbb{I}_{\mathbb{R}_+^2}(x, y)$$

where  $X \sim \text{Exp}(\alpha)$ .

**Option 2:**

1. **CDF:**

$$\begin{aligned} F_X(x) &= \lim_{y \rightarrow \infty} F_{X,Y}(x, y) = 1 - e^{-\alpha x} \\ f_X(x) &= \alpha e^{-\alpha x} \cdot \mathbb{I}_{\mathbb{R}_+}(x) \end{aligned}$$

The marginal of  $Y \Rightarrow Y \sim \text{Exp}(\beta)$ .

2. **PDF:**

$$f_{Y|X}(y|x) = \frac{f_{X,Y}(x, y)}{f_X(x)} = \beta e^{-\beta y} \sim \text{Exp}(\beta) \Rightarrow \text{Independent random variables}$$

3. **PMF:**

$$P(Y \leq y \mid X = x) = \int_{-\infty}^y f_{Y|X}(y|x) dy$$

**N.B.** indicator function over the set of all real numbers greater than zero

$$\mathbb{I}_{\mathbb{R}^+}(x) = \begin{cases} 1, & \text{if } x \in \mathbb{R}^+, \\ 0, & \text{if } x \notin \mathbb{R}^+. \end{cases}$$

## Independence

$$X \perp\!\!\!\perp Y \iff$$

- $F_{X|Y}(x|y) = F_X(x) \rightarrow F_{X,Y}(x,y) = F_X(x)F_Y(y)$
- $P_{X|Y}(x|y) = P_X(x) \rightarrow P_{X,Y}(x,y) = P_X(x)P_Y(y)$
- $f_{X|Y}(x|y) = f_X(x) \rightarrow f_{X,Y}(x,y) = f_X(x)f_Y(y)$

## More Generally

$X_1, \dots, X_p$  collected in a  $p$ -dimensional random vector  $\mathbf{X} = \begin{pmatrix} X_1 \\ \vdots \\ X_p \end{pmatrix}$  are independent

if and only if:

$$F_{\mathbf{X}}(\mathbf{x}) = \prod_{j=1}^p F_{X_j}(x_j)$$

This implies a random sample from a target population  $F_X$ .

## Generalization: Multivariate Case

For a random vector  $\vec{X} = [X_1, \dots, X_p]$ , the joint CDF is:

$$F_{\vec{X}}(\vec{x}) = P(X_1 \leq x_1, \dots, X_p \leq x_p)$$

The random variables  $X_1, \dots, X_p$  are independent if and only if:

$$F_{\vec{X}}(x_1, \dots, x_p) = \prod_{j=1}^p F_{X_j}(x_j)$$

The joint PDF (if continuous) is:

$$f_{\vec{X}}(x_1, \dots, x_p) = \prod_{j=1}^p f_{X_j}(x_j)$$

## Random Sampling

A random sample  $\{X_1, \dots, X_n\}$  is said to be independent and identically distributed (i.i.d.) from a population model  $F_X$  if:

$$F_{X_1, \dots, X_n}(x_1, \dots, x_n) = \prod_{i=1}^n F_X(x_i)$$

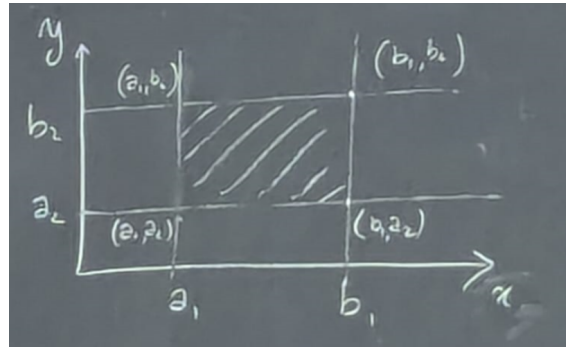
This implies that the data generating process (DGP) is probabilistic, the underlying population model  $F_X$  is only partially known (assumption) and we get information by sampling (independently) observations from it..



## Probability of Rectangles

For a bivariate random vector, the probability of a rectangle  $[a_1, b_1] \times [a_2, b_2]$  is:

$$P(a_1 \leq X \leq b_1 \wedge a_2 \leq Y \leq b_2) = F_{X_1, X_2}(b_1, b_2) - F_{X_1, X_2}(a_1, b_2) - F_{X_1, X_2}(b_1, a_2) + F_{X_1, X_2}(a_1, a_2)$$



## 4.4 Histogram and ECDF: 11/10

### Definitions

- **Histogram (Hist):** Learns an unknown density from data.
- **Empirical Cumulative Distribution Function (ECDF):** Learns an unknown cumulative distribution function (CDF) from data.

### Setup

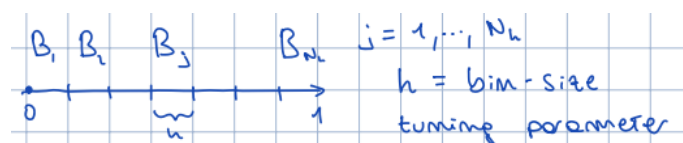
Consider random variables  $X_1, \dots, X_n$  independently sampled from a distribution  $F_X$ , where  $F_X$  is partially unknown. Each  $X_i \in \mathbb{R}$  is supported on  $[0, 1]$ , and represents a continuous random variable.

$$\hat{F}_X(x) = \prod_{i=1}^n F_X(x_i)$$

### 4.4.1 Histogram

We divide the interval  $[0, 1]$  into  $N_h$  bins:  $B_1, B_2, B_3, \dots, B_{N_h}$ , where each bin has a bin-size  $h$  (a tuning parameter).

**Step 1** Divide the range of data into bins  $B_j$ .



**Step 2** For each bin  $B_j$ , evaluate:

- (i) Count how many  $X_i$  fall into that bin.
- (ii) Compute the relative frequency of observations in each bin.
- (iii) Calculate the density by dividing by  $h$ , the size of the bin.

### Adaptive Binning

To estimate densities like  $f_X(x)$ , adaptive binning can be used based on qualitative assumptions about the density function.

$$\hat{f}_h(x) = \sum_{j=1}^{N_h} \mathbb{I}_{B_j}(x) \cdot \frac{\hat{P}_j}{h}$$

where

$$\hat{P}_j = \frac{\#X_i \text{ in } B_j}{n} = \frac{\sum_{i=1}^n \mathbb{I}_{B_j}(X_i)}{n}$$

## 4.4.2 Empirical Cumulative Distribution Function (ECDF)

**Step 1** Consider the empirical distribution of the data.

**Step 2** Evaluate the CDF of this empirical distribution:

$$\hat{F}_n(x) = \frac{1}{n} \sum_{i=1}^n \mathbb{I}(X_i \leq x)$$

Note that for ECDF, there are no tuning parameters. The CDF is a simple, constrained function, whereas the density can be unconstrained and require tuning.

### Indicator Function

The indicator function  $\mathbb{I}_A(x)$  is used to indicate membership in a set  $A$ :

$$\mathbb{I}_A(x) = \begin{cases} 1, & \text{if } x \in A, \\ 0, & \text{if } x \notin A. \end{cases}$$

## 4.5 Mixed Case: Lesson 16/10

### Example

Consider a random vector  $\begin{bmatrix} X \\ W \end{bmatrix}$  where  $W$  is discrete and  $X$  is a generic  $p$ -dimension random vector (continuous).

- The joint distribution of this vector is given by:

$$P(X, W) = P(X|W)P(W)$$

Where the conditional probability  $P(X|W)$  represents the subpopulation model while  $P(W)$  is the marginal PMF of  $W$ , indicating the likelihood of each subpopulation within the overall population .

- To marginalize  $X$  over  $W$ , we have:

$$\sum_{w \in YW} P(X, W) = \sum_{w \in YW} P(X|W)P(W)$$

- This is also known as a finite mixture model (FMM):

$$\mathcal{W} = \{1, \dots, K\}$$

$$\pi_w = \mathbb{P}(W = w)$$

$$p(x) = \sum_{j=1}^K p(x | j)\pi_j$$

where  $\pi_j$  represents the mixing proportion for each sub-population.

The mixing proportions must satisfy the condition:

$$\sum_{j=1}^n \pi_j = 1, \quad \pi_j \geq 0$$

This ensures that the model is properly normalized, and each component weight is non-negative.

### 4.5.1 Gaussian Mixture Models (GMM)

Gaussian Mixture Models (GMM) are a specific type of finite mixture model where each component is a Gaussian distribution.

For the unidimensional case:

$$P(X) = \sum_{j=1}^k N(X|\mu_j, \sigma_j^2)\pi_j$$

where  $N(X|\mu_j, \sigma_j^2)$  represents the Gaussian distribution for component  $j$ , with mean  $\mu_j$  and variance  $\sigma_j^2$ .

**Multidimensional Case** In the multidimensional case,  $X$  is a vector, and each component follows a multivariate Gaussian distribution:

$$P(\mathbf{X}) = \sum_{j=1}^k N(\mathbf{X}|\boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)\pi_j$$

where  $\boldsymbol{\mu}_j$  is the mean vector and  $\boldsymbol{\Sigma}_j$  is the covariance matrix for component  $j$ .

## 4.5.2 Scale Mixture of Gaussians

We want to remove the assumption of a *finite* mixture.

Let  $\begin{pmatrix} X \\ \Lambda \end{pmatrix}$  be a random vector with the joint distribution given by:

$$p(x, \lambda) = p(x \mid \lambda)p(\lambda)$$

where:

$$p(x \mid \lambda) = \mathcal{N}\left(x \mid \mu, \frac{1}{\lambda}\right), \quad p(\lambda) = \text{Gamma}\left(\frac{\lambda}{2}, \frac{\lambda}{2}\right)$$

where  $\lambda > 0$  is a positive scale parameter and  $\lambda$  is a fixed hyperparameter.  
Also  $\mu$  is  $= 0$  and  $\frac{1}{\lambda} = \sigma^2$   
Note that the marginal distribution must take values in  $\mathbb{R}^+$ .

**Marginal Distribution of  $X$  is:**

$$p(x) = \int_0^\infty \mathcal{N}\left(x \mid 0, \frac{1}{\lambda}\right) \text{Gamma}\left(\lambda \mid \frac{\lambda}{2}, \frac{\lambda}{2}\right) d\lambda$$

Where:

- $\mathcal{N}\left(x \mid 0, \frac{1}{\lambda}\right) : \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{x^2}{2\sigma^2}} \propto \sqrt{\lambda} e^{-\frac{\lambda}{2}x^2}$
- $\text{Gamma}\left(\lambda \mid \frac{\lambda}{2}, \frac{\lambda}{2}\right) d\lambda : \propto \lambda^{\frac{\nu}{2}-1} e^{-\lambda\frac{\nu}{2}}$

Substituting the explicit forms of the Normal and Gamma distributions:

$$p(x) \propto \int_0^\infty \lambda^{\frac{1}{2}} e^{-\frac{\lambda x^2}{2}} \lambda^{\frac{\nu}{2}-1} e^{-\lambda\frac{\nu}{2}} d\lambda$$

Simplify to:

$$p(x) \propto \int_0^\infty \lambda^{\frac{\nu+1}{2}-1} e^{-\lambda\frac{(x^2+\nu)}{2}} d\lambda$$

Substituting:

$$t = \lambda \cdot \text{const.}, \quad \Rightarrow \quad \lambda = \text{const.} \cdot t, \quad \Rightarrow \quad d\lambda = \text{const.} \cdot dt$$

$$\int_0^\infty (\text{const.} \cdot t)^{\frac{\nu+1}{2}-1} e^{-t(\text{const.})} dt$$

Using the Gamma function  $\Gamma(s) = \int_0^\infty t^{s-1} e^{-t} dt$ :

$$p(x) \propto (\text{const.})^{\frac{\nu+1}{2}-1} \cdot \Gamma\left(\frac{\nu+1}{2}\right)$$

$$p(x) \propto (\text{const.})^{\frac{\nu+1}{2}-1} \left(1 + \frac{x^2}{\nu}\right)^{-\frac{\nu+1}{2}}$$

This corresponds to a **Student-t distribution**.

So, as  $|x| \rightarrow \infty$ ,  $p(x) \rightarrow 0$  with a polynomial rate

### 4.5.3 Discrete Observations + Cont' Latent Variables

Consider a random vector  $\begin{pmatrix} X \\ \Theta \end{pmatrix}$  with:

$$p(x, \theta) = p(x \mid \theta)p(\theta)$$

Assume:

$$p(x \mid \theta) = \text{Binom}(x \mid \theta, m)$$

where  $p(x, \theta)$  is the probability of success, and  $p(\theta) \sim \text{Uniform}(0, 1)$ .

**The marginal distribution of  $X$  is:**

$$p(x) = \int_0^1 p(x \mid \theta)p(\theta)d\theta = \int_0^1 \binom{m}{x} \theta^x (1 - \theta)^{m-x} d\theta$$

where:

$$\binom{m}{x} = \frac{x!(m-x)!}{m!}, \quad x \in \{0, 1, \dots, m\}$$

since  $x, m \in \mathbb{N}$ , and  $p(x \mid \theta)$  is Binomial.

**Generalization with Beta Distribution** If we assume:

$$p(\theta) = \text{Beta}(\theta \mid \alpha, \beta)$$

then:

$$p(x) = \int_0^1 \binom{m}{x} \theta^x (1 - \theta)^{m-x} \cdot \alpha \theta^{\alpha-1} (1 - \theta)^{\beta-1} d\theta$$

$$p(x) = \binom{m}{x} \int_0^1 \theta^{x+\alpha-1} (1 - \theta)^{(m-x+\beta)-1} d\theta$$

#### Beta-Binomial vs Binomial

**Beta-Binomial:** Model for # of successes in  $m$  Bernoulli trials with unknown/random probability of success  $\Theta$ .

**Binomial:** # of successes in fixed Bernoulli trials with the same probability of success  $\theta$  known.

# Chapter 5

## Expectation e Variance: 5

### 5.1 Introduction to Expectation: 23/10

**Expectation**, also called the mean, is a summary measure that provides a numeric representation of a random variable's central tendency. It is defined for random variables and random vectors, both in discrete and continuous cases.

For a random variable  $X$  with cumulative distribution function (CDF)  $F_X$ :

- **General Definition:**

$$\mathbb{E}(X) = \int x dF_X(x) = \begin{cases} \sum_x x \cdot p_X(x) & \text{if } X \text{ is discrete with PMF } p_X(\cdot), \\ \int x \cdot f_X(x) dx & \text{if } X \text{ is continuous with PDF } f_X(\cdot) \end{cases} \quad (5.1)$$

assuming that the sum/integral is well-defined (i.e.,  $\int |x| dF_X(x) < +\infty$ ).

#### 5.1.1 Properties of Expectation

1. **Dependency on Distribution:** The expectation  $\mathbb{E}(X)$  depends solely on the distribution  $F_X$ , and not the random variable itself. Different random variables may have the same expectation if they share the same distribution.
2. **Comparison with Median:**
  - Median and expectation are generally not equal.
  - Median splits the support into two equal halves, making it more robust to outliers and heavy tails.
  - The expectation acts as a balance point, and can be significantly affected by outliers (e.g., for distributions with heavy tails).

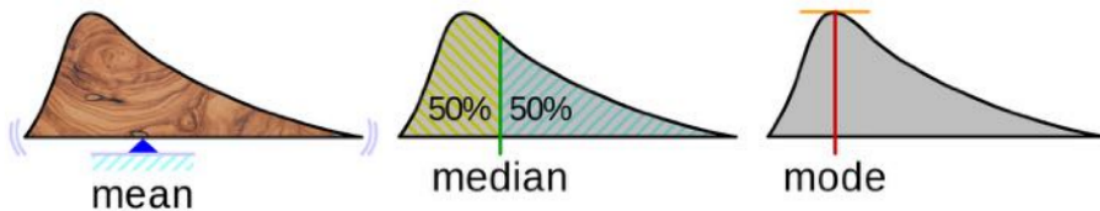


Figure 5.1: Comparison between averages

3. **Symmetry:** For symmetric unimodal distributions:

- If the distribution is symmetric,  $\mathbb{E}(X) = \text{median}(X)$ .
- For a right-skewed distribution:  $\text{median} < \mathbb{E}(X)$ .
- For a left-skewed distribution:  $\text{median} > \mathbb{E}(X)$ .

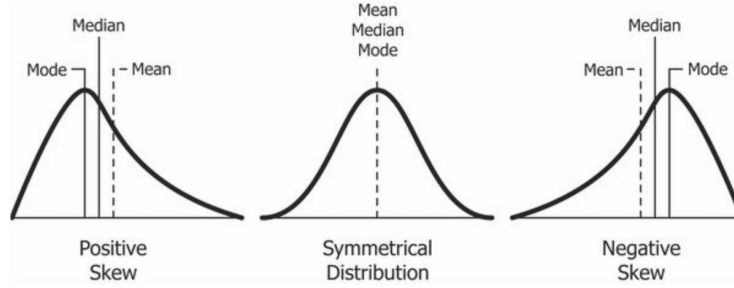


Figure 5.2: Mean and Median for Skewed Distribution

4. **Existence:** The expectation may not exist for some random variables, particularly those with heavy tails (e.g., Cauchy distribution). For instance, the Cauchy distribution has a probability density function given by

$$f_X(x) = \frac{1}{\pi(x^2 + 1)}, \quad \text{for } x \in \mathbb{R}, \quad (5.2)$$

It's a valid PDF. However, due to the heavy tails, the expected value:

$$\mathbb{E}(X) = \int_{-\infty}^{\infty} \frac{x}{\pi(x^2 + 1)} dx, \quad (5.3)$$

diverges to infinity. This means that  $\mathbb{E}(X)$  is not defined.

5. **Equivariance:** The expectation does not always maintain the same relationship after a transformation of the random variable. Specifically, for a transformed variable  $Y = r(X)$ :

- The expectation  $\mathbb{E}(Y)$  is equal to  $r(\mathbb{E}(X))$  only if the transformation  $r(x)$  is affine, meaning it has the form  $r(x) = \alpha + \beta x$ . In general, for non-linear transformations, this relationship does not hold.
- The median is equivariant under one-to-one (bijective) transformations, meaning the median of a transformed variable  $Y = r(X)$  can be obtained by applying the transformation  $r$  to the median of  $X$ .

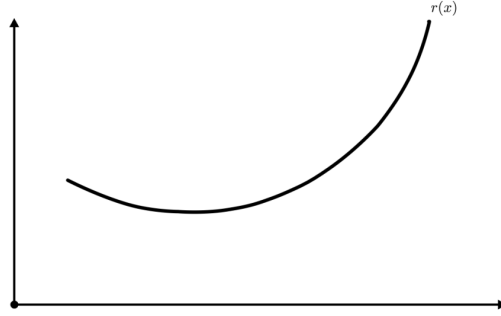
6. **Law of the Lazy Statistician (LLS):** The Law of the Lazy Statistician allows us to calculate the expected value of a function of a random variable directly from the distribution of the original random variable. Specifically, let  $X \sim F_X$  and  $Y = r(X)$  for some deterministic function  $r(\cdot)$ , then:

$$\mathbb{E}(Y) = \int r(x) dF_X(x) = \int r(x) f_X(x) dx \quad (5.4)$$

This property holds for both continuous and discrete random variables and applies to general transformations, regardless of linearity.

7. **Jensen's Inequality:** Jensen's inequality describes the relationship between the expectation of a convex function and the function applied to the expectation. Specifically, for a convex function  $r(x)$ , we have:

$$r(\mathbb{E}(X)) \leq \mathbb{E}(r(X)) \quad (5.5)$$

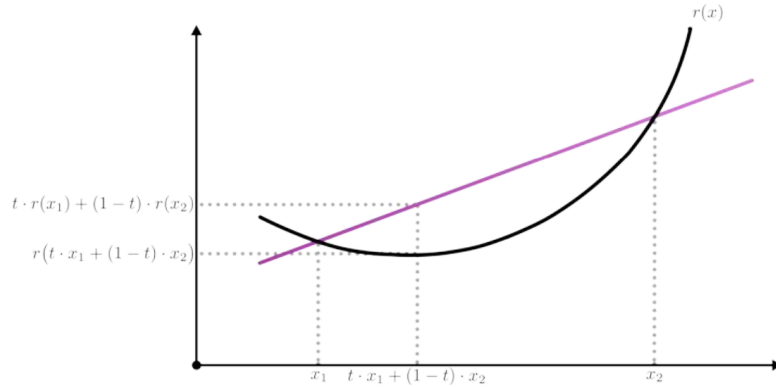


If  $r(\cdot)$  is a convex function:

- geometrically it means that the function's graph has a positive (upward) curvature at every point.
- for a twice-differentiable function  $r(\cdot)$ , the convexity condition implies  $r''(x) > 0$  for all  $x$ .

Consider two points  $x_1, x_2 \in S$  and  $t \in [0, 1]$ .

A function  $r : \mathbb{R} \rightarrow \mathbb{R}$  is convex if:  $r(tx_1 + (1-t)x_2) \leq tr(x_1) + (1-t)r(x_2)$



Imagine a discrete R. var  $X$  supported on  $\{x_1, x_2\}$  with probabilities  $P(X = x_1) = t$  and  $P(X = x_2) = 1 - t$ . The expectation of  $X$  is given by:

$$\mathbb{E}(X) = tx_1 + (1-t)x_2 \quad (5.6)$$

Consider a convex function  $r(\cdot)$ . The expectation of  $r(X)$  is:

$$\mathbb{E}(r(X)) = t \cdot r(x_1) + (1-t) \cdot r(x_2) \quad (5.7)$$

By the definition of convexity:

$$r(\mathbb{E}(X)) \leq \mathbb{E}(r(X)) \quad (5.8)$$



8. **Gap between Expectation and Median:** The gap between expectation and median is bounded by the standard deviation:  
 let  $\mu$  be the expectation of our population model (assuming it exists!) and let  $m$  be its median, then:

$$|\mu - m| = |\mathbb{E}(X) - m| = \mathbb{E}(|X - m|) \leq \mathbb{E}(|X - \mu|) \leq \mathbb{E}\left(\sqrt{(X - \mu)^2}\right) = \sqrt{\mathbb{E}((X - \mu)^2)} = \sigma_X$$

### 5.1.2 Prediction under Perfect Information

In statistics, a common problem is predicting the value of a R. var  $X$  with mean  $\mu$  and variance  $\sigma^2$ . To make the best prediction before observing the value, we use a:

#### Decisional Framework

- We select a **loss function**  $L(\cdot, \cdot)$  that measures the cost or error of predicting a candidate value  $d$ . Common loss functions include:
  - **Square Loss**  $((X - d)^2)$  (Regression) emphasizes larger deviations, making the mean the ideal estimate to minimize such variations.
  - **Absolute Loss**  $(|X - d|)$  (Robust Regression) provides robustness against outliers, making the median the optimal solution.
  - **0-1 Loss**  $(\mathbb{I}(X = d))$  (Classification) applies to classification problems, focusing on minimizing the probability of misclassification.

**Mean Squared Error (MSE)** When the **square loss** is used as the loss function, the goal becomes minimizing the **Mean Squared Error** (MSE):

$$d^* = \arg \min_d \mathbb{E}[(X - d)^2]. \quad (5.9)$$

The value that minimizes this expression is the mean of  $X$ ,  $\mu = \mathbb{E}(X)$ .

**Mean Absolute Error (MAE) and the Median** If we use the **absolute loss** function, the goal is to minimize the **Mean Absolute Error** (MAE):

$$d^* = \arg \min_d \mathbb{E}[|X - d|]. \quad (5.10)$$

The value that minimizes this expression is the median of  $X$ .

## 5.2 Variance and Its Properties: 25/10

**Variance** measures the spread of a random variable, quantifying how far values typically are from the mean.

- The variance of a R. var  $X$  with mean  $\mu_X = \mathbb{E}(X)$  is defined as:

$$\mathbb{E}((X - \mu_X)^2) =_{\text{LLS}} \int (x - \mu_X)^2 dF_X(x). \quad (5.11)$$

- Standard deviation is the square root of variance:  $\sigma_X = \sqrt{\text{Var}(X)}$ .

We can forge a more robust counterpart to the variance by simply changing the represented (using median for instance)

## Basic Properties

1. **Non-Negativity and Existence:** For any R. var  $X$ ,  $\text{Var}(X) \geq 0$ . If  $X$  is a bounded random variable, then  $\text{Var}(X)$  must exist and be finite.
2. **Degeneracy:** For a random variable  $X$ , if  $\sigma_X^2 = 0$ , then  $X$  is almost surely equal to its mean:

$$\sigma_X^2 = 0 \iff X \text{ a.s.} = \mathbb{E}(X). \quad (5.12)$$

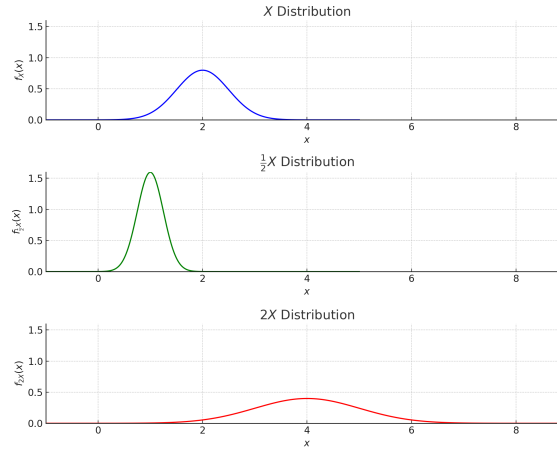
This means that  $X$  takes the value  $\mu_X$  with probability 1 (almost surely), indicating no variability.

3. **Alternative Calculation of Variance:**

$$\begin{aligned} \text{Var}(X) &= \mathbb{E}[(X - \mu_X)^2] \\ &= \mathbb{E}(X^2 + \mu_X^2 - 2X\mu_X) \\ &= \mathbb{E}(X^2) + \mu_X^2 - 2\mu_X\mathbb{E}(X) \\ &= \mathbb{E}(X^2) - \mu_X^2 \end{aligned}$$

4. **Linearity Property:** For a linear transformation  $Y = \alpha + \beta X$ :

$$\begin{aligned} \text{Var}(a + bX) &= b^2 \text{Var}(X) \\ &= \mathbb{E}[(a + bX - a - b\mu_X)^2] \\ &= b^2 \mathbb{E}[(X - \mu_X)^2] \\ &= b^2 \text{Var}(X) \end{aligned}$$



Scaling a random variable scales its variance by the square of the scaling factor, while adding a constant shift  $\alpha$  does not affect the variance.

## 5. Variance: Topic 30/10

The variance of a random variable  $X$  is defined as:

$$\text{Var}(X) = \mathbb{E}[(X - \mu_X)^2] = \int (x - \mu_X)^2 dF_X(x)$$

For the variance of the sum of independent variables  $X_i$ :

$$\text{Var}\left(\sum_{i=1}^M X_i\right) \stackrel{?}{=} \sum_{i=1}^M \text{Var}(X_i)$$

where  $M = 2$  and  $\mathbb{E}(X_i) = 0$ .

## 6. Variance of Sum of Two Variables

For two random variables  $X_1$  and  $X_2$ :

$$\text{Var}(X_1 + X_2) = \mathbb{E}_X[(X_1 + X_2)^2] = \mathbb{E}_X(X_1^2 + X_2^2 + 2X_1X_2)$$

which simplifies to:

$$\begin{aligned} &= \mathbb{E}_{X_1}(X_1^2) + \mathbb{E}_{X_2}(X_2^2) + 2\mathbb{E}_{X_1, X_2}(X_1X_2) \\ &= \text{Var}(X_1) + \text{Var}(X_2) + 2\mathbb{E}_{X_1, X_2}(X_1X_2) \end{aligned}$$

The term  $\mathbb{E}_{X_1, X_2}(X_1X_2)$  is referred to as the **first cross moment**.

## 7. General Case

In general, for independent random variables  $X_i$ :

$$\mathbb{E}\left[\prod_{i=1}^M X_i\right] = \prod_{i=1}^M \mathbb{E}(X_i)$$

In this specific case,  $\mathbb{E}(X_1X_2)$  represents the **covariance** because  $\mathbb{E}(X_1) = \mathbb{E}(X_2) = 0$ . Due to independence:

$$\mathbb{E}(X_1X_2) = 0$$

## 8. Variance of the Sum of Independent Variables

Therefore, for independent variables  $X_i$ :

$$\text{Var}\left(\sum_{i=1}^M X_i\right) = \sum_{i=1}^M \text{Var}(X_i) = np(1-p)$$

## 9. First Cross Moment (Covariance)

The first cross moment, or covariance, between  $X_1$  and  $X_2$  is defined as:

$$\mathbb{E}_{X_1, X_2}(X_1X_2) = \iint x_1 \cdot x_2 dF_{X_1, X_2}(x_1, x_2)$$

If  $X_1$  and  $X_2$  are independent, then:

$$F_{X_1, X_2}(x_1, x_2) = F_{X_1}(x_1) \cdot F_{X_2}(x_2)$$

This implies:

$$\mathbb{E}_{X_1, X_2}(X_1X_2) = \int x_1 dF_{X_1}(x_1) \int x_2 dF_{X_2}(x_2) = \mathbb{E}_{X_1}(X_1) \cdot \mathbb{E}_{X_2}(X_2)$$

## 10. Moments of a Distribution:

$$\mu^{(j)} = \mathbb{E}(X^j)$$

The  $j$ -th moment of a random variable  $X$  exists if  $\mathbb{E}(|X|^j) < +\infty$ . If the  $k$ -th moment is finite, then all moments of order  $j < k$  are also finite.

**Examples:**

- $\mathbb{E}(X^3)$ : Related to skewness (asymmetry of the distribution).
- $\mathbb{E}(X^4)$ : Related to kurtosis (tail heaviness of the distribution).

**Moment Condition for Existence**

$$\mathbb{E}(|X|^j) = \int_{-\infty}^{+\infty} |x|^j dF_X(x) = \int_{|x| \leq 1} |x|^j dF_X(x) + \int_{|x| > 1} |x|^j dF_X(x)$$

Using a bound:

$$\mathbb{E}(|X|^j) \leq 1 + \mathbb{E}(|X|^k), \quad \text{for } j < k.$$

### 5.2.1 Law of Large Numbers (LLN)

The **Law of Large Numbers** (LLN) states that as the number of independent and identically distributed (i.i.d.) samples  $X_1, \dots, X_n$  from a distribution  $F_X$  increases, the sample mean  $\bar{X}_n$  converges to the true expectation  $\mu_X = \mathbb{E}(X)$ :

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{a.s.} \mu_X \quad \text{as } n \rightarrow +\infty. \quad (5.13)$$

This ensures that as the sample size grows, the sample mean becomes a consistent estimator of the population mean.

### 5.2.2 Conditional Expectation

In general, let  $X$  be a  $p$ -dimensional random vector:

$$X = \begin{bmatrix} X_1 \\ \vdots \\ X_p \end{bmatrix} \sim F_X$$

The expectation  $\mathbb{E}(X)$  is given by:

$$\mathbb{E}(X) = \begin{bmatrix} \mathbb{E}(X_1) \\ \vdots \\ \mathbb{E}(X_p) \end{bmatrix}$$

Suppose we split  $X$  into two parts:

$$X = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$$

where  $X_1$  is a  $k$ -dimensional random vector and  $X_2$  is a  $p - k$  dimensional random vector. The conditional expectation  $\mathbb{E}(X_1|X_2)$  is of interest.

**Example: Air Pollution**

Given information  $X$ ,  $\mathbb{E}(Y|X)$  is the best predictor of  $Y$  under the square loss.

### 5.2.3 Random Vector $(X, Y)$

Let  $(X, Y) \sim F_{X,Y}$  be a joint distribution, we can describe it using:

- - Discrete case:  $P_{X,Y}(x, y)$ , the joint PMF.
- - Continuous case:  $f_{X,Y}(x, y)$ , the joint PDF.

#### Derived Distributions

**Marginals:**

$$f_X(x) = \int f_{X,Y}(x, y) dy$$

$$f_Y(y) = \int f_{X,Y}(x, y) dx$$

**Conditionals:**

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}$$

$$f_{Y|X}(y|x) = \frac{f_{X,Y}(x, y)}{f_X(x)}$$

#### Conditional Expectations

$$\mathbb{E}(Y|X = x) = \int_{\mathbb{R}} y \cdot f_{Y|X}(y|x) dy$$

$$\mathbb{E}(X|Y = y) = \int_{\mathbb{R}} x \cdot f_{X|Y}(x|y) dx$$

#### Changing Notation and Interpretation

The notation  $\mathbb{E}(Y|X) \approx \mathbb{E}(Y|X = x)$  is used to capture the variability of  $(X, Y)$ , with  $X$  capturing the information about  $Y$ . This conditional expectation represents a new random variable function of  $X$ .

$$\mathbb{E}_X [\mathbb{E}_{Y|X}(Y|X)] = \int_{S_X} f_X(x) \left( \int_{S_Y} y f_{Y|X}(y|x) dy \right) dx$$

Expanding:

$$\begin{aligned} &= \int_{S_X} \int_{S_Y} y \cdot \frac{f_{X,Y}(x, y)}{f_X(x)} dy f_X(x) dx \\ &= \int_{S_Y} y \cdot f_Y(y) dy = \mathbb{E}(Y) \end{aligned}$$

### 5.2.4 Law of Total Expectation

The Law of Total Expectation states:

$$\mathbb{E}(Y) = \mathbb{E}_X [\mathbb{E}_{Y|X}(Y|X)]$$

This is linked to the Law of Total Probability.

In terms of probability:

$$P(E) = \sum_{j=1}^k P(E|H_j)P(H_j)$$

Consider the indicator variable  $Y = 1(E)$ , where  $X$  is a discrete random variable with  $k$  values in the support, one for each  $H_j$ . Then  $Y \sim \text{Bernoulli}(P(E))$  with:

$$\mathbb{E}_Y(Y) = P(E)$$

$$P(X = j) = P(H_j)$$

Expectation equals probability:

$$\mathbb{E}(Y) = \mathbb{E}_X [\mathbb{E}_{Y|X}(Y|X)] = P(E) = \sum_{j=1}^k P(E|H_j)P(H_j)$$

### 5.2.5 Law of Total Variance

The variance of  $Y$  is given by:

$$\text{Var}(Y) = \mathbb{E}(Y^2) - \mathbb{E}(Y)^2 = \mathbb{E}_X [\mathbb{E}_{Y|X}(Y^2|X)] - (\mathbb{E}_X [\mathbb{E}_{Y|X}(Y|X)])^2$$

which expands to:

$$\begin{aligned} &= \mathbb{E}_X [\text{Var}_{Y|X}(Y|X) + (\mathbb{E}_{Y|X}(Y|X))^2] - (\mathbb{E}_X [\mathbb{E}_{Y|X}(Y|X)])^2 \\ &= \mathbb{E}_X [\text{Var}_{Y|X}(Y|X)] + \text{Var}_X [\mathbb{E}_{Y|X}(Y|X)] \end{aligned}$$

This is the **Law of Total Variance**.

# Chapter 6

## Dependency and correlation

### 6.1 Overview

Consider a random vector in  $\mathbb{R}^n$  representing a collection of random variables. These variables may exhibit different types of dependency. Our objective is to create simple numerical summaries that can capture or respond to these dependencies.

#### Types of Dependency

- **Linear Dependency** - correlation between variables with linear relationships.
- **Monotonic Dependency** - dependency where  $g(X)$  is monotonic, reflecting a consistent relationship direction.
- **Arbitrary Dependency Structure** - non-linear, complex dependencies modelled by functions  $g_i$  for some  $i$ .

### 6.2 Linear Dependency

Let  $X$  and  $Y$  be random variables with means  $\mu_i$  and standard deviation  $\sigma_i$ , we can define:

- The covariance between  $X$  and  $Y$  as  $\text{Cov}(X, Y)$ , defined as:

$$\sigma_{X,Y} = \mathbb{E}((X - \mathbb{E}(X))(Y - \mathbb{E}(Y))) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y)$$

- The Pearson linear correlation coefficient  $\rho_{X,Y}$  between the two variables defined as:

$$\rho_{X,Y} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)}\sqrt{\text{Var}(Y)}} = \frac{\sigma_{X,Y}}{\sigma_X\sigma_Y}$$

The covariance and the correlation coefficient measure how strong the linear relationship between two random variables  $X$  and  $Y$  actually is. This can be interpreted as a measure of collinearity.

### 6.2.1 Standardization and correlation

To design a dependency measure, the following properties should not impact the dependency measure :

- **Marginal Average**
- **Marginal Variability**

In other words, we should cleaning the original random variable from this marginal effects before obtaining any summary. In order to reduce this marginal effects, the random variable  $X$  is standardized as:

$$Z = \frac{X - \mathbb{E}[X]}{\sigma}$$

where  $\mathbb{E}[Z] = 0$  and  $\text{Var}(Z) = 1$ . For two standardized random variables  $Z_1$  and  $Z_2$ , the covariance simplifies as follows:

$$\text{Cov}(Z_1, Z_2) = \mathbb{E}[Z_1 Z_2] = \rho_{X,Y}$$

The Pearson coefficient takes values between  $-1$  and  $1$ , where:

- $\rho = 1$  or  $\rho = -1$  indicates a perfect linear relationship defined as:

$$Y = \left[ \mu_Y - \left( \frac{\sigma_Y}{\sigma_X} \right) \mu_X \right] + \left( -\frac{\sigma_Y}{\sigma_X} \right) X$$

- $\rho = 0$  implies no linear relationship (uncorrelated).

If the two variables are independent the correlation is equal to zero, but the reversal is not always true.

### 6.2.2 Geometry of Correlation

For all pair of standardized random variables  $Z_X$  and  $Z_Y$ , the correlation can be defined as:

$$\rho_{X,Y} = \cos(\theta) = \frac{\vec{Z}_X \cdot \vec{Z}_Y}{\|\vec{Z}_X\| \|\vec{Z}_Y\|}$$

where  $\theta$  is the angle between  $\vec{Z}_X$  and  $\vec{Z}_Y$ . So we can say that:

- positive correlation  $\leftrightarrow$  acute angle between corresponding vectors
- redzero correlation  $\leftrightarrow$  right angle between corresponding vectors
- negative correlation  $\leftrightarrow$  obtuse angle between corresponding vectors

Moreover:

- Suppose that  $X$  and  $Y$ , and  $Y$  and  $Z$ , are positively correlated random variables with correlations  $\rho_{X,Y} > 0$  and  $\rho_{Y,Z} > 0$  respectively, and that  $\rho_{X,Y}^2 + \rho_{Y,Z}^2 > 1$ , then  $X$  and  $Z$  are positively correlated.
- For any three random variables, the correlations must satisfy the following inequality:

$$\rho_{X,Y}^2 + \rho_{Y,Z}^2 + \rho_{X,Z}^2 - 1 \leq 2 \cdot \rho_{X,Y} \cdot \rho_{Y,Z} \cdot \rho_{X,Z}$$



### 6.2.3 Notation

#### Variance–Covariance Matrix

For a *generic* random vector  $\mathbf{X} = [X_1, \dots, X_p]^T \in \mathbb{R}^p$ , we collect their variances  $\{\sigma_i\}_i$  and covariances  $\{\sigma_{i,j}\}_{(i,j)}$  into a single  $(p \times p)$  symmetric and positive semi-definite matrix:

$$\text{Cov}(\mathbf{X}) = \Sigma = \begin{bmatrix} \sigma_1^2 & \sigma_{1,2} & \dots & \sigma_{1,p} \\ \sigma_{1,2} & \sigma_2^2 & \dots & \sigma_{2,p} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{1,p} & \sigma_{2,p} & \dots & \sigma_p^2 \end{bmatrix} \rightsquigarrow \Lambda = \Sigma^{-1} \text{ is called the precision matrix.}$$

#### Spectral Decomposition and Total Variance

- Let  $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p \geq 0$  be the ordered **eigenvalues** of  $\Sigma$  and  $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_p\}$  be the set of corresponding (orthonormal) **eigenvectors**.
- Then  $\Sigma = UDU^T$  where  $D = \text{diag}(\lambda_1, \dots, \lambda_p)$  and  $U = [\mathbf{u}_1 | \dots | \mathbf{u}_p]$ .
- Consequently, by the **circularity** of the trace and orthogonality of  $U$ , we get

$$\sum_{j=1}^p \sigma_j^2 = \text{trace}(\Sigma) = \text{trace}(UDU^T) = \text{trace}(U^TUD) = \text{trace}(D) = \sum_{j=1}^p \lambda_j$$

#### Correlation Matrix

Closely related to the variance–covariance matrix is the **correlation matrix**, the matrix of Pearson correlation coefficients between each of the random variables in the random vector  $\mathbf{X}$ :

$$\rho(\mathbf{X}) = R = (\text{diag}(\Sigma))^{-\frac{1}{2}} \Sigma (\text{diag}(\Sigma))^{-\frac{1}{2}}$$

where  $\text{diag}(\Sigma)$  is the diagonal matrix of the variances of  $X_i$ .

# Chapter 7

## Graphical Models [7b]

### 7.0.1 Motivation

Consider a discrete random vector  $X \in \mathbb{R}^D$  with joint distribution  $p(X)$ . Using the chain rule of probability, we can express the joint distribution as:

$$p(x_1, x_2, \dots, x_D) = p(x_D | x_1, \dots, x_{D-1}) p(x_{D-1} | x_1, \dots, x_{D-2}) \dots p(x_1) \quad (7.1)$$

However, as the number of variables  $D$  increases, this representation becomes computationally expensive. The number of free parameters required grows exponentially with  $D$ , making exact inference impossible.

To address this issue, we introduce conditional independence assumptions, which reduce the number of dependencies and allow for a more efficient representation of the joint probability distribution.

### 7.0.2 Conditional Independence and Factorization

Conditional independence is a fundamental concept in graphical models. Two random variables  $X$  and  $Y$  are conditionally independent given  $Z$  if:

$$p(X, Y | Z) = p(X | Z) p(Y | Z) \quad (7.2)$$

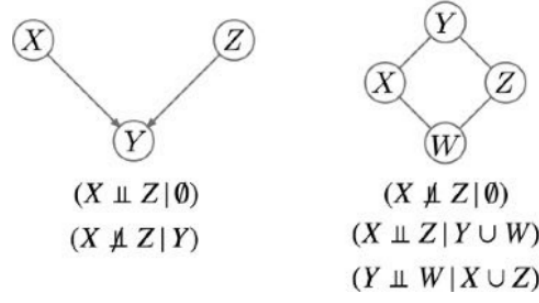
which is denoted as  $X \perp Y | Z$ .

This property significantly reduces the number of parameters needed to describe the distribution, as we do not need to consider direct dependencies between variables that are conditionally independent.

### 7.0.3 Types of Graphical Models

Graphical models can be classified into two main categories:

- **Directed Graphical Models (Bayesian Networks):** Represent dependencies using directed edges and are useful for modeling causal relationships.
- **Undirected Graphical Models (Markov Random Fields):** Represent dependencies using undirected edges and are commonly used in spatial models and computer vision.



## 7.1 Directed Graphical Models (Bayesian Networks)

A Directed Acyclic Graph (DAG) is a directed graphical model that encodes dependencies through directed edges.

### 7.1.1 Terminology

Let  $G = (V, E)$  be a DAG with nodes  $V$  representing random variables and directed edges  $E$  indicating dependencies.

- **Parent:** Nodes that have direct edges pointing to a variable.
- **Child:** Nodes that receive an edge from a variable.
- **Root:** A node with no parents.
- **Leaf:** A node with no children.
- **Ancestors:** The set of all predecessors of a node.
- **Descendants:** The set of all successors of a node.
- **Neighbors:** The neighbors of a node is the set of all immediately connected nodes
- **Cycle:** A cycle is a series of nodes such that we can get back to where we started by following edges.
- **Path:** A path  $s \rightarrow \dots \rightarrow t$  is a series of directed edges leading from  $s$  to  $t$ .

### 7.1.2 Markov Condition in Bayesian Networks

We define the **ordered Markov property** or simply **Markov property** as the assumption:

*“Each node in the graph depends only on its **immediate parents**, not all predecessors in the ordering.”*

Mathematically, this is expressed as:

$$(X_d \perp X_{\text{pred}(d) \setminus \text{pa}(d)} \mid X_{\text{pa}(d)}) \quad \forall d \in \{1, \dots, D\}. \quad (7.3)$$

where  $\text{pa}(d)$  represents the parents of node  $d$ , and  $\text{pred}(d)$  are its predecessors in the ordering.

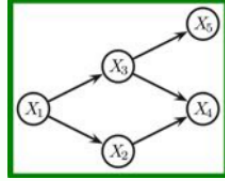
### 7.1.3 Graph Induced Factorization

Under the Markov condition, the joint probability distribution factorizes as:

$$p(x_{1:D} \mid G) = \prod_{d=1}^D p(x_d \mid x_{\text{pa}(d)}). \quad (7.4)$$

This equation holds only if the conditional independence (CI) assumptions encoded by the graph  $G$  are correct.

Graph induced factorization: Example

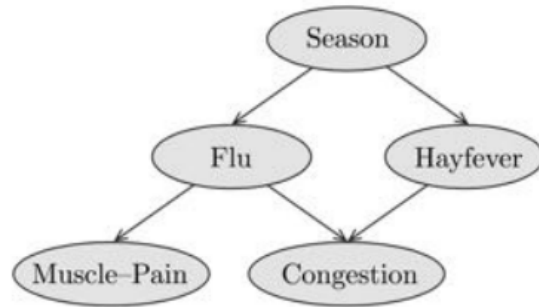


Factorization encoded by the DAG:

$$\begin{aligned} p(\mathbf{x}_{1:5}) &\stackrel{\text{CR}}{=} p(x_1)p(x_2|x_1)p(x_3|x_1, \cancel{x_2}) \\ &\quad p(x_4|\cancel{x_1}, x_2, x_3)p(x_5|\cancel{x_1}, \cancel{x_2}, x_3, \cancel{x_4}) \\ &\stackrel{\text{DAG}}{=} p(x_1)p(x_2|x_1)p(x_3|x_1) \\ &\quad p(x_4|x_2, x_3)p(x_5|x_3) \end{aligned}$$

## Bayesian networks

### Graph Representation



### Independencies

- (Flu  $\perp\!\!\!\perp$  Hayfever | Season)
- (Congestion  $\perp\!\!\!\perp$  Season | Flu, Hayfever)
- (Muscle-Pain  $\perp\!\!\!\perp$  Hayfever, Congestion | Flu)
- (Muscle-Pain  $\perp\!\!\!\perp$  Congestion | Flu)

### Factorization

$$\begin{aligned} p(S, F, H, C, M) &= p(S)p(F|S) \\ &\quad p(H|S)p(C|F, H)p(M|F) \end{aligned}$$

## 7.2 d-Separation

In Bayesian Networks, **d-separation** is a criterion used to determine whether two sets of nodes are conditionally independent given a third set.

A set of variables  $A$  is **d-separated** from another set  $B$  by a third set  $C$  if every path between  $A$  and  $B$  is blocked given  $C$ . A path is considered **blocked** if at least one of the following conditions holds:

To fully understand d-separation, consider the following cases:

**Chain Structure:**  $X \rightarrow Y \rightarrow Z$

In a chain structure, information flows through the middle variable:

- If  $Y$  is **not observed**, then  $X$  and  $Z$  are **dependent**.
- If  $Y$  is **observed**, then  $X$  and  $Z$  become **independent**.

This is because knowing  $Y$  provides all necessary information about  $Z$ , making additional knowledge of  $X$  unnecessary.

**Fork Structure:**  $X \leftarrow Y \rightarrow Z$

In this case, both  $X$  and  $Z$  share a common cause  $Y$ :

- If  $Y$  is **not observed**, then  $X$  and  $Z$  are **dependent** since they are both influenced by  $Y$ .
- If  $Y$  is **observed**, then  $X$  and  $Z$  become **independent**, as  $Y$  accounts for their correlation.

**Collider Structure:**  $X \rightarrow Y \leftarrow Z$  (**Explaining Away**)

Here,  $X$  and  $Z$  influence a common effect  $Y$ :

- If  $Y$  is **not observed**, then  $X$  and  $Z$  are **independent**.
- If  $Y$  is **observed**, then  $X$  and  $Z$  become **dependent** (explaining away effect).

This effect occurs because, knowing that  $Y$  has occurred, observing  $X$  makes  $Z$  less likely and vice versa.

# Chapter 8

## Probability Inequalities [10]

Probability inequalities provide upper bounds for probabilities that are difficult to compute directly.

### Gaussian Tail Inequality

**Theorem 1 (Gaussian Tail Inequality)** Let  $X \sim \mathcal{N}(0, 1)$ . Then, for any  $\epsilon > 0$ :

$$P(|X| > \epsilon) \leq \frac{2e^{-\epsilon^2/2}}{\epsilon}. \quad (8.1)$$

If  $X_1, \dots, X_n \sim \mathcal{N}(0, 1)$ , then:

$$P(|\bar{X}_n| > \epsilon) \leq \frac{2}{\sqrt{n}\epsilon} e^{-n\epsilon^2/2} \leq e^{-n\epsilon^2/2}. \quad (8.2)$$

**Proof:** The probability density function of  $X$  is:

$$\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}. \quad (8.3)$$

Applying integration bounds and properties of the normal distribution leads to the stated inequality.

### Markov's Inequality

**Theorem 2 (Markov's Inequality)** Let  $X$  be a non-negative random variable with finite expectation  $E(X)$ . Then, for any  $t > 0$ :

$$P(X > t) \leq \frac{E(X)}{t}. \quad (8.4)$$

**Proof:** Since  $X \geq 0$ :

$$E(X) = \int_0^\infty xp(x)dx \geq \int_t^\infty xp(x)dx \geq t \int_t^\infty p(x)dx = tP(X > t). \quad (8.5)$$

### Chebyshev's Inequality

**Theorem 3 (Chebyshev's Inequality)** Let  $X$  be a random variable with mean  $\mu = E(X)$  and variance  $\sigma^2 = \text{Var}(X)$ . Then, for any  $t > 0$ :

$$P(|X - \mu| \geq t) \leq \frac{\sigma^2}{t^2}. \quad (8.6)$$

This inequality states that the probability of  $X$  deviating from its mean by at least  $t$  is at most  $\sigma^2/t^2$ .

### Proof of Chebyshev's Inequality

Applying Markov's inequality to the squared deviation  $(X - \mu)^2$ :

$$P(|X - \mu| \geq t) = P((X - \mu)^2 \geq t^2) \leq \frac{E[(X - \mu)^2]}{t^2} = \frac{\sigma^2}{t^2}. \quad (8.7)$$

- Setting  $t = 2\sigma$ :  $P(|X - \mu| \geq 2\sigma) \leq \frac{1}{4}$ .
- Setting  $t = 3\sigma$ :  $P(|X - \mu| \geq 3\sigma) \leq \frac{1}{9}$ .

This special cases illustrate that large deviations from the mean are increasingly unlikely, even in distributions that are not necessarily normal.

### Example: Application to Bernoulli Random Variables

Consider  $X_1, \dots, X_n \sim \text{Bernoulli}(p)$  with mean  $E(X_i) = p$  and variance  $\text{Var}(X_i) = p(1 - p)$ . Define the sample mean:

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i. \quad (8.8)$$

Using the variance formula:

$$\text{Var}(\bar{X}_n) = \frac{p(1 - p)}{n}, \quad (8.9)$$

applying Chebyshev's inequality gives:

$$P(|\bar{X}_n - p| \geq \epsilon) \leq \frac{p(1 - p)}{n\epsilon^2} \leq \frac{1}{4n\epsilon^2}, \quad (8.10)$$

where the last bound follows because  $p(1 - p) \leq 1/4$  for all  $p \in [0, 1]$ .

This result highlights that the sample mean  $\bar{X}_n$  concentrates around  $p$  as  $n$  increases, a foundational concept in probability and statistics.

## 8.1 Convergence

### 8.1.1 Convergence in Probability

A sequence of random variables  $\{X_n\}$  converges in probability to a random variable  $X$  if, for every  $\epsilon > 0$ :

$$\lim_{n \rightarrow \infty} P(|X_n - X| \geq \epsilon) = 0. \quad (8.11)$$

This means that as  $n$  grows, the probability that  $X_n$  deviates significantly from  $X$  becomes arbitrarily small.

**Example:** Let  $X_n = p + \frac{1}{n}$ , where  $p$  is a constant. Then,

$$P(|X_n - p| \geq \epsilon) = 0, \quad \forall \epsilon > 0, n > 1/\epsilon. \quad (8.12)$$

Thus,  $X_n$  converges in probability to  $p$ .

### 8.1.2 Almost Sure Convergence

A sequence of random variables  $\{X_n\}$  converges almost surely to a random variable  $X$  if:

$$P\left(\lim_{n \rightarrow \infty} X_n = X\right) = 1. \quad (8.13)$$

This means that for almost every sample point, the sequence  $X_n$  eventually gets arbitrarily close to  $X$  and stays there.

**Example:** Consider a fair coin flip sequence  $X_n$  where:

$$X_n = \begin{cases} 1 & \text{if the first } n \text{ flips are all heads,} \\ 0 & \text{otherwise.} \end{cases} \quad (8.14)$$

Since  $P(X_n = 1) = \frac{1}{2^n}$ , we see that  $X_n$  converges almost surely to 0.

### Relationship Between Convergence Modes

The different types of convergence are related as follows:

- Almost sure convergence  $\Rightarrow$  Convergence in probability.
- Convergence in probability does **not** necessarily imply almost sure convergence.

### 8.1.3 Laws of Large Numbers (LLN)

#### Weak Law of Large Numbers (WLLN)

The WLLN states that the sample mean of i.i.d. random variables converges in probability to the expected value:

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{P} E(X). \quad (8.15)$$

**Proof:** Using Chebyshev's inequality:

$$P(|\bar{X}_n - E(X)| \geq \epsilon) \leq \frac{\text{Var}(X)}{n\epsilon^2} \rightarrow 0 \text{ as } n \rightarrow \infty. \quad (8.16)$$



## Strong Law of Large Numbers (SLLN)

The SLLN strengthens this result, ensuring almost sure convergence:

$$\bar{X}_n \xrightarrow{a.s.} E(X). \quad (8.17)$$

This implies that with probability 1, the sample mean eventually stabilizes around  $E(X)$  and does not fluctuate indefinitely.

## 8.1.4 The Central Limit Theorem (CLT)

The Central Limit Theorem (CLT) is one of the most fundamental results in probability theory. It states that, under mild conditions, the sum (or average) of a large number of independent random variables is approximately normally distributed, regardless of the original distribution of the variables.

### Statement of the CLT

Let  $X_1, X_2, \dots, X_n$  be a sequence of i.i.d. random variables with mean  $\mu$  and variance  $\sigma^2$ . Define the sample mean:

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i. \quad (8.18)$$

Then, the standardized version of  $\bar{X}_n$ :

$$Z_n = \frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \quad (8.19)$$

converges in distribution to a standard normal random variable  $Z \sim N(0, 1)$ :

$$\lim_{n \rightarrow \infty} P(Z_n \leq z) = \Phi(z), \quad (8.20)$$

where  $\Phi(z)$  is the cumulative distribution function (CDF) of the standard normal distribution.

### Interpretation

The CLT implies that for large  $n$ , the sample mean  $\bar{X}_n$  is approximately normally distributed:

$$\bar{X}_n \approx N\left(\mu, \frac{\sigma^2}{n}\right). \quad (8.21)$$

This result allows us to use normal distribution approximations for many statistical inference problems.

### Example: Application to Sample Means

Consider  $X_1, X_2, \dots, X_n \sim \text{Uniform}(0, 1)$ . Here,  $E(X_i) = 1/2$  and  $\text{Var}(X_i) = 1/12$ . By the CLT, for large  $n$ :

$$\frac{\sqrt{n}(\bar{X}_n - 1/2)}{\sqrt{1/12}} \approx N(0, 1). \quad (8.22)$$

For  $n = 100$ , we approximate:

$$P(0.45 \leq \bar{X}_{100} \leq 0.55) \approx P(-1.73 \leq Z \leq 1.73) \approx 0.958. \quad (8.23)$$

This allows for normal approximations of sample means in practice.

# Chapter 9

## Statistical Inference [11]

### 9.1 Statistical Inference

Statistical inference refers to the process of drawing conclusions about a population from a sample of data. This process assumes:

1. A **parametric** or **nonparametric** model for our measurements, denoted as:

$$X \sim F_X \tag{9.1}$$

where  $F_X$  represents the underlying distribution.

2. We acquire information in the form of an independent and identically distributed (iid) random sample:

$$X_n = \{X_1, X_2, \dots, X_n\} \tag{9.2}$$

The **grand goal** of statistical inference is to learn about population features or parameters of interest using data.

#### 9.1.1 Inferential Paradigms: Frequentist vs. Bayesian

Statistical inference is broadly divided into two inferential paradigms:

##### Frequentist Inference

In the frequentist approach:

- The population feature (parameter) is fixed but unknown.
- Inferential procedures are random because they depend on the random sample.
- The performance of an estimator is evaluated under repeated sampling from the population.
- The **sampling distribution** of an estimator plays a crucial role in inference.

## Bayesian Inference

In the Bayesian framework:

- The population feature (parameter) is considered **random** because it is unknown.
- Inferential procedures depend on prior beliefs and observed data.
- The posterior distribution of the parameter, given the observed data, is the main object of interest:

$$p(\theta|X_n) \propto p(X_n|\theta)p(\theta) \quad (9.3)$$

where  $p(\theta)$  is the prior,  $p(X_n|\theta)$  is the likelihood, and  $p(\theta|X_n)$  is the posterior distribution.

### Key differences:

The fundamental distinctions between frequentist and Bayesian inference are summarized in Table 9.1.

| Aspect                       | Frequentist                 | Bayesian           |
|------------------------------|-----------------------------|--------------------|
| Probability Interpretation   | Limiting relative frequency | Degree of belief   |
| Population Parameter         | Fixed but unknown           | Random variable    |
| Observable Data ( $X$ )      | Random variable             | Random variable    |
| Probability Statements About | Procedures                  | Parameters         |
| Frequency Guarantees         | Yes, by construction        | No, not in general |
| Use of Bayes' Theorem        | To build procedures         | To update beliefs  |

Table 9.1: Comparison of Frequentist and Bayesian Inference

## 9.1.2 Parametric vs. Nonparametric Inference

### Parametric Inference

The parametric approach assumes that the distribution  $F_X(\cdot)$  belongs to a known family of distributions characterized by a finite set of parameters:

$$F_X(x) = F_X(x|\theta), \quad \theta \in \Theta \subset \mathbb{R}^k \quad (9.4)$$

where  $\theta$  represents the parameters (e.g., mean and variance in a normal distribution). Parametric methods rely on a substantial reduction in the complexity of the inference problem.

### Nonparametric Inference

In contrast, nonparametric methods avoid strong assumptions about the functional form of  $F_X(\cdot)$ . They estimate:

- The cumulative distribution function (CDF):  $F_X(x) = P(X \leq x)$ .

- Statistical functionals, such as the mean  $E[X]$ .
- Density functions:  $f_X(x) = F'_X(x)$ .
- Regression functions:  $r(X) = E[Y|X]$ .

Unlike parametric methods, nonparametric approaches require specialized techniques, such as kernel density estimation or spline regression, to manage their infinite-dimensional nature.

### 9.1.3 Main Learning Problems in Statistical Inference

The primary objectives of statistical inference fall into three categories:

- **Point Estimation:** Estimating unknown parameters with a single value.
- **Set Estimation:** Constructing confidence intervals or credible intervals for parameters.
- **Hypothesis Testing:** Testing specific claims about parameters or distributions.

## 9.2 Point Estimation

Point estimation is the process of using sample data to provide a single, best-guess estimate of an unknown population parameter. Given a random sample:  $X_n = \{X_1, X_2, \dots, X_n\}$  from a distribution  $F_X(\cdot)$ , our goal is to estimate a population quantity of interest  $\theta$ .

In parametric inference,  $\theta$  is a fixed parameter (e.g., mean or variance), while in nonparametric inference,  $\theta$  may represent a more complex feature such as an entire cumulative distribution function (CDF).

### Definition and Notation

- A point estimator  $\hat{\theta}_n$  is a function of the observed data:

$$\hat{\theta}_n = \hat{\theta}(X_1, X_2, \dots, X_n) \quad (9.5)$$

- The realized value of the estimator after observing data is called the **point estimate**.
- The probability distribution of  $\hat{\theta}_n$  is called the **sampling distribution**.
- The standard deviation of  $\hat{\theta}_n$  is referred to as the **standard error**:

$$\text{se}(\hat{\theta}_n) = \sqrt{\text{Var}(\hat{\theta}_n)} \quad (9.6)$$

- How far on average the r.v.  $\hat{\theta}_n$  is from its inferential target  $\theta$  is called the **bias**, denoted

$$\text{bias}(\hat{\theta}_n) = \theta - \mathbb{E}(\hat{\theta}_n).$$

### 9.2.1 Risk Function and Expectation of Loss

Since  $\hat{\theta}_n$  is random (as a function of  $X_n$ ), the loss function is also random. To evaluate an estimator, we consider the expected loss, also called the **risk function**:

$$R_\theta(\hat{\theta}_n) = E[L(\theta, \hat{\theta}_n)] \quad (9.7)$$

Using the Law of the Lazy Statistician (LLS), the risk can be computed as:

$$R_\theta(\hat{\theta}_n) = \int L(\theta, u) dG_n(u), \quad \text{if } G_n(\cdot) \text{ is known} \quad (9.8)$$

where  $G_n(\cdot)$  is the sampling distribution of  $\hat{\theta}_n$ . When  $G_n(\cdot)$  is unknown, integration is performed over the full data distribution  $F_X(\cdot)$ .

### Quadratic Loss and Mean Squared Error (MSE)

For numeric parameters  $\theta \in \Theta \subseteq \mathbb{R}$ , the quality of an estimator is often assessed using the **mean squared error (MSE)**:

$$\text{MSE}(\hat{\theta}_n) = E[(\hat{\theta}_n - \theta)^2] \quad (9.9)$$

For vector-valued parameters  $\theta \in \mathbb{R}^k$ , we use:

$$\text{MSE}(\hat{\theta}_n) = \sum_{j=1}^k E[(\hat{\theta}_{n,j} - \theta_j)^2] \quad (9.10)$$

MSE represents the expected squared distance of  $\hat{\theta}_n$  from  $\theta$ .

### Bias-Variance Decomposition of MSE

MSE can be decomposed as:

$$\text{MSE}(\hat{\theta}_n) = \text{Var}(\hat{\theta}_n) + \text{Bias}^2(\hat{\theta}_n) \quad (9.11)$$

where:

- **Variance**: Measures how much  $\hat{\theta}_n$  fluctuates across different samples.
- **Bias**: Measures how far the expected value of  $\hat{\theta}_n$  is from the true  $\theta$ :

$$\text{Bias}(\hat{\theta}_n) = E[\hat{\theta}_n] - \theta \quad (9.12)$$

An estimator is **unbiased** if  $\text{Bias}(\hat{\theta}_n) = 0$ . However, some biased estimators may still be preferable due to lower MSE.

### 9.2.2 Consistency and Asymptotics

A desirable property of an estimator is **consistency**, meaning that as the sample size grows,  $\hat{\theta}_n$  converges to  $\theta$ :

$$\hat{\theta}_n \xrightarrow{P} \theta \quad \text{as } n \rightarrow \infty \quad (9.13)$$

For MSE consistency:

- $\text{Bias}(\hat{\theta}_n) \rightarrow 0$  as  $n \rightarrow \infty$ .
- $\text{Var}(\hat{\theta}_n) \rightarrow 0$  as  $n \rightarrow \infty$ .

By the Central Limit Theorem (CLT), many estimators are asymptotically normal:

$$\frac{\hat{\theta}_n - \theta}{\text{se}(\hat{\theta}_n)} \xrightarrow{d} N(0, 1) \quad (9.14)$$

where  $\text{se}(\hat{\theta}_n)$  is the standard error of  $\hat{\theta}_n$ .

## 9.3 Nonparametric Point Estimation

Nonparametric estimation aims to infer population characteristics without assuming a specific parametric model for the underlying distribution. Given a random sample:

$$X_n = \{X_1, X_2, \dots, X_n\} \quad (9.15)$$

from an unknown distribution  $F_X(x) = P(X \leq x)$ , our goal is to estimate either:

- The entire cumulative distribution function (CDF)  $F_X(\cdot)$ .
- A functional parameter derived from  $F_X(\cdot)$  (e.g., quantiles, density function).

Unlike parametric methods, nonparametric estimation does not assume a finite set of parameters but instead estimates the function directly from data.

### 9.3.1 Empirical Distribution Function (EDF)

The most fundamental nonparametric estimator for the CDF  $F_X(x)$  is the **empirical distribution function** (EDF), denoted as  $\hat{F}_n(x)$ . The EDF is defined as:

$$\hat{F}_n(x) = \frac{1}{n} \sum_{i=1}^n I(X_i \leq x) \quad (9.16)$$

where  $I(X_i \leq x)$  is an indicator function that equals 1 if  $X_i \leq x$  and 0 otherwise.

#### Properties of the EDF

For a fixed  $x$ , the random variables  $I(X_i \leq x)$  are independent and identically distributed (iid) Bernoulli variables with parameter:

$$p_x = P(X_i \leq x) = F_X(x) \quad (9.17)$$

This leads to the following properties:

- **Unbiasedness:**

$$E[\hat{\theta}_n^{(0)}] = \frac{nE(W_i^{x_0})}{n} = F_X(x_0) = \theta^{(0)} \quad (9.18)$$

Hence,  $\hat{\theta}_n^{(0)}$  is an unbiased estimator of  $\theta^{(0)}$ .

- **Variance:**

$$\text{Var}(\hat{\theta}_n^{(0)}) = \frac{1}{n} \text{Var}(W_i^{x_0}) = \frac{F_X(x_0)(1 - F_X(x_0))}{n} \quad (9.19)$$

which decreases as  $n \rightarrow \infty$ .

- **Mean Squared Error (MSE):**

$$\text{MSE}(\hat{\theta}_n^{(0)}) = \text{Var}(\hat{\theta}_n^{(0)}) \rightarrow 0 \quad \text{as } n \rightarrow \infty \quad (9.20)$$

This confirms that  $\hat{\theta}_n^{(0)}$  is **consistent**.

- **Law of Large Numbers (LLN):** By the LLN, we already knew:

$$\hat{\theta}_n^{(0)} \xrightarrow{P} F_X(x_0) = \theta^{(0)} \quad (9.21)$$

- **Asymptotic Normality:** By the Central Limit Theorem (CLT),

$$\sqrt{n} \left( \hat{\theta}_n^{(0)} - \theta^{(0)} \right) \xrightarrow{d} N(0, \theta^{(0)}(1 - \theta^{(0)})) \quad (9.22)$$

- The standardized version follows:

$$\hat{\theta}_n^{(0)} \sim N_1(\theta^{(0)}, \hat{se}^2) \quad (9.23)$$

where the estimated standard error is given by:

$$\hat{se}^2 = \frac{\hat{\theta}_n^{(0)}(1 - \hat{\theta}_n^{(0)})}{n} \quad (9.24)$$

### 9.3.2 Uniform Estimation (Estimating the Entire CDF)

Instead of estimating  $F_X(x)$  at a single point, we estimate the entire function  $F_X(\cdot)$ . This is a more challenging problem because we require a uniform guarantee over all  $x$  values:

$$\theta = F_X(\cdot) \in \mathcal{F} = \{\text{space of all CDFs}\} = \Theta. \quad (9.25)$$

The natural estimator is the empirical distribution function (EDF):

$$\hat{\theta}_n = \hat{F}_n(\cdot). \quad (9.26)$$

To analyze the properties of  $\hat{F}_n(\cdot)$ , we introduce a stronger, uniform version of the Law of Large Numbers (LLN), known as the **Glivenko–Cantelli Theorem**, sometimes called the **Fundamental Theorem of Statistics**.

[Glivenko–Cantelli (GC)] For any true cumulative distribution function  $F_X(x)$ , the empirical distribution function  $\hat{F}_n(x)$  satisfies:

$$\sup_x |\hat{F}_n(x) - F_X(x)| \xrightarrow{a.s.} 0 \quad \text{as } n \rightarrow \infty. \quad (9.27)$$

This means that the EDF converges to the true CDF **uniformly** almost surely, ensuring that we control the worst-case deviation across all  $x$ . This result strengthens pointwise consistency by guaranteeing that the entire function  $\hat{F}_n(x)$  approximates  $F_X(x)$  well across all values.

## 9.4 Introduction to Set Estimation

In contrast to point estimation, where a single value is used to estimate an unknown parameter, set estimation aims to provide a range of values that likely contain the true parameter. The most common approaches include:

- **Confidence Intervals** (Frequentist Approach): Intervals constructed from sample data that, under repeated sampling, contain the true parameter with a specified probability.
- **Credible Intervals** (Bayesian Approach): Posterior probability intervals representing the degree of belief about the true parameter's location.
- **Tolerance Sets**: Intervals or regions capturing a specified proportion of the population.

### 9.4.1 Confidence Intervals

A **confidence interval** at level  $(1 - \alpha)$  is a random set computed from the given data:

$$C_n(\alpha) = [L(X_n), U(X_n)] \quad (9.28)$$

where  $X_n = \{X_1, \dots, X_n\} \stackrel{\text{iid}}{\sim} F_X(\cdot)$ . The confidence interval satisfies:

$$P_{F_X}(\theta \in C_n(\alpha)) \geq 1 - \alpha \quad \text{uniformly over } \theta. \quad (9.29)$$

where  $1 - \alpha$  is the confidence level, typically chosen as 95% or 99%.

Confidence intervals can be constructed using:

- **Normal Approximation**: Based on the Central Limit Theorem (CLT), if an estimator  $\hat{\theta}_n$  is approximately normal,

$$\hat{\theta}_n \pm z_{1-\alpha/2} \cdot \text{SE}(\hat{\theta}_n) \quad (9.30)$$

where  $z_{1-\alpha/2}$  is the standard normal quantile and  $\text{SE}(\hat{\theta}_n)$  is the standard error.

- **Bootstrap Methods**: Resampling techniques used to construct CIs when normality assumptions do not hold.
- **T-Distribution**: When estimating means with small sample sizes, the Student's t-distribution provides improved interval estimates.

A common misconception is that a 95% CI means that there is a 95% probability that  $\theta$  is within the interval. The correct frequentist interpretation is:

If we repeatedly sample and compute CIs, 95% of such intervals will contain  $\theta$ .

In the Bayesian framework, a **credible interval**  $[L_n, U_n]$  satisfies:

$$P(L_n \leq \theta \leq U_n | X_n) = 1 - \alpha \quad (9.31)$$

where the probability is computed from the posterior distribution  $p(\theta | X_n)$ .

**Equal-Tailed Interval**: Satisfies

$$P(\theta < L_n | X_n) = \frac{\alpha}{2}, \quad P(\theta > U_n | X_n) = \frac{\alpha}{2}. \quad (9.32)$$



## 9.5 Statistical Functionals

In many statistical applications, we are not directly interested in estimating the cumulative distribution function (CDF)  $F_X(\cdot)$ , but rather some specific feature of the data-generating process, such as the mean or variance. These quantities are often expressed as **statistical functionals**.

A **statistical functional**  $g(F_X)$  is any transformation of  $F_X(\cdot)$ . Some typical examples include:

- **Mean:**  $\mu = E(X) = \int x dF_X(x)$ .
- **Variance:**  $\sigma^2 = \text{Var}(X) = \int (x - \mu)^2 dF_X(x)$ .
- **Median:**  $m = F_X^{-1}(1/2)$ .
- **Covariance:**  $\sigma_{X,Y} = E(XY) - E(X)E(Y)$ .

A functional of the form  $g_r(F_X) = \int r(x) dF_X(x)$  is called **linear** because it satisfies the property:

$$g_r(c_1 F_X + c_2 G_X) = c_1 g_r(F_X) + c_2 g_r(G_X). \quad (9.33)$$

### 9.5.1 Plug-in Principle

A common approach to estimating a statistical functional is the **plug-in principle**, which replaces the unknown CDF  $F_X$  with the empirical CDF (ECDF)  $\hat{F}_n$ .

#### Plug-in Estimator

The plug-in estimator of a functional  $\theta = g(F_X)$  is defined as:

$$\hat{\theta}_n = g(\hat{F}_n). \quad (9.34)$$

Since the ECDF  $\hat{F}_n(x)$  is discrete, putting mass  $1/n$  at each  $X_i$ , for any linear functional  $g_r(F_X)$ , the plug-in estimator simplifies to:

$$\hat{\theta}_n = g_r(\hat{F}_n) = \int r(x) d\hat{F}_n(x) = \frac{1}{n} \sum_{i=1}^n r(X_i). \quad (9.35)$$

This means that for linear functionals, the plug-in estimator reduces to a simple sample average.

#### Examples

##### Example 1: The Mean

$$\hat{\mu}_n = \int x d\hat{F}_n(x) = \frac{1}{n} \sum_{i=1}^n X_i = \bar{X}_n. \quad (9.36)$$

##### Example 2: The Variance

$$\hat{\sigma}_n^2 = \frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}_n^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2. \quad (9.37)$$

### Example 3: Skewness

$$\hat{\kappa}_n = \frac{\frac{1}{n} \sum_{i=1}^n (X_i - \hat{\mu}_n)^3}{\hat{\sigma}_n^3}. \quad (9.38)$$

### Example 4: Sample Correlation

$$\hat{\rho}_n = \frac{\sum_{i=1}^n (X_i - \bar{X}_n)(Y_i - \bar{Y}_n)}{\sqrt{\sum_{i=1}^n (X_i - \bar{X}_n)^2} \sqrt{\sum_{i=1}^n (Y_i - \bar{Y}_n)^2}}. \quad (9.39)$$

## Properties of Plug-in Estimators

The plug-in estimator  $\hat{\theta}_n$  inherits properties from  $\hat{F}_n$ .

- **Consistency:** The **Dvoretzky-Kiefer-Wolfowitz (DKW)** and **Glivenko-Cantelli** theorems ensure that  $\hat{F}_n(\cdot)$  converges to  $F_X(\cdot)$ .
- **Asymptotic Normality:** Under mild conditions,  $\hat{\theta}_n$  is asymptotically normal:

$$\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{d} N(0, \sigma^2). \quad (9.40)$$

However, the plug-in estimator is not always consistent.

For density estimation:

$$\theta = g(F_X) = \frac{d}{dx} F_X(x). \quad (9.41)$$

The plug-in estimator  $\hat{\theta}_n = \frac{d}{dx} \hat{F}_n(x)$  is a poor estimate since  $\hat{F}_n$  is discrete, leading to a step function with undefined derivatives.

# Chapter 10

## Bootstrap

The **bootstrap** is a powerful resampling technique introduced by Bradley Efron to approximate the sampling distribution of an estimator when the theoretical distribution is unknown or difficult to compute. The method is useful for:

- Estimating standard errors,
- Constructing confidence intervals,
- Assessing bias.

Unlike classical approaches relying on parametric assumptions, the bootstrap uses the observed data itself to infer properties of the estimator.

### 10.0.1 Bootstrap Resampling Procedure

Given an observed sample  $X_n = \{X_1, X_2, \dots, X_n\}$ , the bootstrap consists of the following steps:

1. Draw a **bootstrap sample**  $X_n^*$  of size  $n$  by sampling with replacement from  $X_n$ .
2. Compute the statistic of interest  $\hat{\theta}_b^* = g(X_n^*)$  for this resampled dataset.
3. Repeat steps 1-2 for  $B$  iterations to obtain  $\{\hat{\theta}_1^*, \dots, \hat{\theta}_B^*\}$ .
4. Use the empirical distribution of the bootstrap estimates to approximate quantities of interest.

### The Bootstrap Analogy

| Concept            | Real World                                      | Bootstrap World                                       |
|--------------------|-------------------------------------------------|-------------------------------------------------------|
| True Distribution  | $F_X(\cdot)$                                    | $\hat{F}_n(\cdot)$ (Empirical Distribution)           |
| Sample/Data        | $\{X_1, \dots, X_n\} \sim F_X(\cdot)$           | $\{X_1^*, \dots, X_n^*\} \sim \hat{F}_n(\cdot)$       |
| Parameter          | $\theta = g(F_X)$                               | $\hat{\theta}_n = g(\hat{F}_n)$                       |
| Estimator          | $\hat{\theta}_n$                                | $\theta_n^* = g(\hat{F}_n^*)$                         |
| Error              | $\hat{\theta}_n - \theta$                       | $\theta_n^* - \hat{\theta}_n$                         |
| Standardized Error | $\frac{\hat{\theta}_n - \theta}{se(\hat{F}_n)}$ | $\frac{\theta_n^* - \hat{\theta}_n}{se(\hat{F}_n^*)}$ |
| Bias               | $E_{F_X}(\hat{\theta}_n) - \theta$              | $E_{\hat{F}_n}(\theta_n^*) - \hat{\theta}_n$          |
| Variance           | $\text{Var}_{F_X}(\hat{\theta}_n)$              | $\text{Var}_{\hat{F}_n}(\theta_n^*)$                  |

In the bootstrap world, everything is **known**. The empirical distribution  $\hat{F}_n(\cdot)$  acts as a surrogate for the true distribution  $F_X(\cdot)$ , and the bootstrap estimate  $\hat{\theta}_n$  serves as an approximation of the true parameter  $\theta$ .

- $\hat{F}_n(\cdot)$  plays the role of the true unknown distribution, and  $\hat{\theta}_n$  plays the role of the true unknown parameter  $\theta$ .
- The sampling distribution of  $\theta_n^*$  or its error  $\theta_n^* - \hat{\theta}_n$  may still be difficult to compute theoretically, but it can be approximated using Monte Carlo resampling.
- Sampling from  $\hat{F}_n(\cdot)$  is equivalent to sampling from a finite population where the “population” consists of the observed data  $\{X_1, \dots, X_n\}$ .
- Since we work **conditionally** on the observed sample, we must resample **with replacement** to ensure the resampled dataset is *i.i.d.*.
- Thus,  $\{X_1^*, \dots, X_n^*\}$  is a bootstrap sample obtained by sampling with replacement from  $\{X_1, \dots, X_n\}$ . This process is called **resampling**.

## 10.1 Bootstrapping Standard Errors

A central concept in frequentist inference is the standard error of an estimator  $\hat{\theta}_n$ , defined as the square root of its variance:

$$SE(\hat{\theta}_n) = \sqrt{\text{Var}(\hat{\theta}_n)}. \quad (10.1)$$

However, direct formulas for standard errors exist mainly for averages and a few other statistics, making the bootstrap a powerful tool for estimating them.

### The Variance of an Estimator

Given a random sample  $\{X_1, \dots, X_n\}$  from  $F_X(\cdot)$ , the variance of an estimator  $\hat{\theta}_n$  is defined as:

$$\text{Var}(\hat{\theta}_n) = \int \dots \int \left( \hat{\theta}_n(x_1, \dots, x_n) - \mu \right)^2 dF_X(x_1) \dots dF_X(x_n), \quad (10.2)$$

where  $\mu = E(\hat{\theta}_n)$ . Computing this directly is often infeasible, requiring approximation methods such as the bootstrap.

#### 10.1.1 Monte Carlo Approximation way

Instead of evaluating the full integral, we approximate the variance numerically by generating bootstrap resamples:

1. Draw  $B$  bootstrap samples  $X_n^*$  by resampling with replacement from the observed data.
2. Compute  $\hat{\theta}_b^*$  for each resampled dataset.

3. Estimate the variance using:

$$\widehat{\text{Var}}(\hat{\theta}_n) = \frac{1}{B} \sum_{b=1}^B \left( \hat{\theta}_b^* - \bar{\theta}^* \right)^2, \quad (10.3)$$

where  $\bar{\theta}^* = \frac{1}{B} \sum_{b=1}^B \hat{\theta}_b^*$ .

The Law of Large Numbers ensures that the approximation improves as  $B \rightarrow \infty$ .

## 10.2 Bootstrap Confidence Intervals

Confidence intervals (CIs) provide a range of values that likely contain the true parameter  $\theta$  with a given probability  $(1-\alpha)$ . Constructing exact confidence intervals often requires knowledge of the true distribution  $F_X(\cdot)$ , which is typically unknown. The bootstrap provides an approximation using resampling methods. Several methods exist for constructing bootstrap confidence intervals, each with different assumptions and levels of accuracy.

### 10.2.1 Normal Approximation Interval

If  $\hat{\theta}_n$  is approximately normal:

$$C_n(\alpha) = \left[ \hat{\theta}_n - z_{\alpha/2} \cdot \text{SE}_{\text{boot}}, \hat{\theta}_n + z_{\alpha/2} \cdot \text{SE}_{\text{boot}} \right], \quad (10.4)$$

where  $z_{\alpha/2}$  is the standard normal quantile and  $\text{SE}_{\text{boot}}$  is the bootstrap standard error.

### 10.2.2 Percentile Bootstrap Interval

The percentile bootstrap CI is constructed by taking the empirical quantiles of the bootstrap distribution:

$$C_n(\alpha) = [Q_{\alpha/2}(\theta_n^*), Q_{1-\alpha/2}(\theta_n^*)]. \quad (10.5)$$

This method assumes that the bootstrap distribution of  $\theta_n^*$  adequately approximates the true sampling distribution.

### 10.2.3 Pivotal Bootstrap Interval

The pivotal bootstrap method utilizes the studentized bootstrap statistic:

$$Z_n^* = \frac{\theta_n^* - \hat{\theta}_n}{\text{SE}_{\text{boot}}}, \quad (10.6)$$

and constructs the confidence interval as:

$$C_n(\alpha) = \left[ 2\hat{\theta}_n - Q_{1-\alpha/2}(Z_n^*), 2\hat{\theta}_n - Q_{\alpha/2}(Z_n^*) \right]. \quad (10.7)$$

This method improves accuracy by correcting skewness in the sampling distribution.

# Chapter 11

## Parametric Inference and the Parametric Bootstrap

### 11.0.1 Parametric Inference

Parametric inference assumes that the data is generated from a known parametric family of distributions  $\mathcal{F} = \{f(x | \theta) : \theta \in \Theta \subseteq \mathbb{R}^k\}$  where  $\theta$  is the parameter of interest. The objective is to estimate  $\theta$  using observed data  $X_1, X_2, \dots, X_n$ .

### 11.1 Method of Moments Estimation

The **Method of Moments** (MM) estimator  $\hat{\theta}_{MM}$  is obtained by matching sample moments with theoretical moments:

$$\mathbb{E}_\theta[g(X)] = \frac{1}{n} \sum_{i=1}^n g(X_i). \quad (11.1)$$

This approach provides a simple but often suboptimal estimator.

### 11.2 Maximum Likelihood Estimation (MLE)

The **likelihood function** is defined as the joint density of the observed data, treated as a function of the parameter:

$$L_n(\theta | X_n) = \prod_{i=1}^n f(X_i | \theta). \quad (11.2)$$

Taking the natural logarithm gives the **log-likelihood function**:

$$\ell_n(\theta | X_n) = \log L_n(\theta | X_n) = \sum_{i=1}^n \log f(X_i | \theta). \quad (11.3)$$

The **Maximum Likelihood Estimator** (MLE) is obtained by maximizing the log-likelihood:

$$\hat{\theta}_{MLE} = \arg \max_{\theta \in \Theta} L_n(\theta | X_n) = \arg \max_{\theta \in \Theta} \ell_n(\theta | X_n). \quad (11.4)$$

In other words, the MLE is the parameter value that maximizes the joint probability of observing the given sample.

## Dropping Constants in Likelihood Maximization

Since the MLE is invariant to multiplicative constants that do not depend on  $\theta$ , we often drop them for simplification. That is, if  $c > 0$  is a constant independent of  $\theta$ , then:

$$\arg \max_{\theta \in \Theta} cL_n(\theta | X_n) = \arg \max_{\theta \in \Theta} L_n(\theta | X_n). \quad (11.5)$$

This property is useful when simplifying the likelihood for mathematical convenience.

## 11.3 Properties of MLE

MLEs have several desirable properties:

1. **Consistency:** The MLE  $\hat{\theta}_{MLE}$  converges to the true parameter  $\theta$  as the sample size  $n$  increases:

$$\hat{\theta}_{MLE} \rightarrow \theta \quad \text{as } n \rightarrow \infty. \quad (11.6)$$

2. **Asymptotic Normality:** The MLE is approximately normal for large  $n$ :

$$\sqrt{n}(\hat{\theta}_{MLE} - \theta) \xrightarrow{d} \mathcal{N}(0, I(\theta)^{-1}), \quad (11.7)$$

where  $I(\theta)$  is the Fisher information matrix.

3. **Equivariance:** If we reparameterize the model using a function  $h(\theta)$ , then the MLE of the transformed parameter is simply  $h(\hat{\theta}_{MLE})$ :

$$\hat{\tau}_{MLE} = h(\hat{\theta}_{MLE}). \quad (11.8)$$

4. **Asymptotic Efficiency:** Among consistent estimators, the MLE achieves the smallest possible variance:

$$\text{Var}(\hat{\theta}_{MLE}) \approx I(\theta)^{-1}. \quad (11.9)$$

## 11.4 Maximum Likelihood Estimators

To obtain approximate standard errors and confidence sets for the parameter of interest, we follow different approaches:

- **Nonparametric Model:**

- Analytic solution via influence functions.
- Computational solution via Nonparametric Bootstrap.

- **Parametric Model:**

- Analytic solution via Fisher Information.
- Computational solution via Parametric Bootstrap.

To simplify, we first analyze the uni-parametric case where  $\Theta \rightarrow \mathbb{R}$  and assume the parametric model is well specified.

### 11.4.1 Asymptotic Normality of MLE

The MLE is asymptotically normal, and its estimated standard error can be computed analytically. We define:

- The **Score Function**: If  $X \sim f(x|\theta)$ , then

$$s(X|\theta) = \frac{\partial}{\partial \theta} \log f(X|\theta). \quad (11.10)$$

- The **Fisher Information** for i.i.d. samples  $X_1, \dots, X_n$ :

$$I_n(\theta) = \sum_{i=1}^n \text{Var}(s(X_i|\theta)), \quad (11.11)$$

where for a single observation, we write:

$$I(\theta) = -\mathbb{E} \left[ \frac{\partial^2}{\partial \theta^2} \log f(X|\theta) \right]. \quad (11.12)$$

The Fisher Information measures how much information a sample carries about the parameter. A higher curvature in the log-likelihood means higher Fisher Information, leading to smaller standard errors.

Let  $\hat{\theta}_n$  be the MLE of  $\theta$ . Then, under regularity conditions:

- For large  $n$ , the sampling distribution of  $\hat{\theta}_n$  is approximately normal:

$$\hat{\theta}_n \sim \mathcal{N} \left( \theta, \frac{1}{I_n(\theta)} \right). \quad (11.13)$$

- The standard error (SE) is approximated by:

$$\text{SE}(\hat{\theta}_n) \approx \frac{1}{\sqrt{I_n(\theta)}}. \quad (11.14)$$

Since  $I_n(\theta)$  depends on the unknown parameter  $\theta$ , we estimate it by plugging in  $\hat{\theta}_n$ :

$$\text{SE}(\hat{\theta}_n) \approx \frac{1}{\sqrt{I_n(\hat{\theta}_n)}}. \quad (11.15)$$

### 11.4.2 Example: Bernoulli Model

Consider i.i.d. samples  $X_1, \dots, X_n \sim \text{Bernoulli}(p)$ . The probability mass function is:

$$f(x|p) = p^x(1-p)^{1-x}, \quad x \in \{0, 1\}. \quad (11.16)$$

The MLE for  $p$  is simply the sample mean:

$$\hat{p}_n = \frac{1}{n} \sum_{i=1}^n X_i. \quad (11.17)$$



We derive the log-likelihood function:

$$\log f(x|p) = x \log p + (1 - x) \log(1 - p). \quad (11.18)$$

Taking the derivative with respect to  $p$  gives the score function:

$$s(X|p) = \frac{X}{p} - \frac{1 - X}{1 - p}. \quad (11.19)$$

Taking the second derivative:

$$-\frac{\partial}{\partial p} s(X|p) = \frac{X}{p^2} + \frac{1 - X}{(1 - p)^2}. \quad (11.20)$$

Computing the Fisher Information:

$$I(p) = \mathbb{E} \left[ -\frac{\partial}{\partial p} s(X|p) \right] = \frac{1}{p(1 - p)}. \quad (11.21)$$

Thus, the estimated standard error of  $\hat{p}_n$  is:

$$\text{SE}(\hat{p}_n) = \sqrt{\frac{\hat{p}_n(1 - \hat{p}_n)}{n}}. \quad (11.22)$$

For large  $n$ , the MLE  $\hat{p}_n$  is approximately normal:

$$\hat{p}_n \sim \mathcal{N} \left( p, \frac{p(1 - p)}{n} \right). \quad (11.23)$$

This allows us to construct confidence intervals for  $p$  easily.

### 11.4.3 Normal Confidence Intervals

#### Recap

A confidence interval at level  $(1 - \alpha)$  is a random set  $C_n(\alpha) = [L_n, U_n]$  such that:

$$P_{F_X}(\theta \in C_n(\alpha)) \geq 1 - \alpha \quad \text{uniformly over } \theta. \quad (11.24)$$

## 11.5 Normal Intervals for the Mean (Asymptotic Pivot)

If  $\hat{\theta}_n$  is asymptotically normal, we assume:

$$\hat{\theta}_n \sim \mathcal{N}(\theta, \text{se}^2(\hat{\theta}_n)). \quad (11.25)$$

Let  $z_{\alpha/2}$  be the quantile:

$$z_{\alpha/2} = -\text{qnorm}(\alpha/2) = \text{qnorm}(1 - \alpha/2). \quad (11.26)$$

Standardizing  $\hat{\theta}_n$  gives:

$$Z = \frac{\hat{\theta}_n - \theta}{\text{se}(\hat{\theta}_n)} \sim \mathcal{N}(0, 1). \quad (11.27)$$

Rearranging, we obtain the **asymptotic confidence interval**:

$$C_n(\alpha) = \left[ \hat{\theta}_n - z_{\alpha/2} \text{se}(\hat{\theta}_n), \hat{\theta}_n + z_{\alpha/2} \text{se}(\hat{\theta}_n) \right]. \quad (11.28)$$

### 11.5.1 The $\delta$ -Method

The previous framework allows us to construct normal confidence intervals for  $\theta$ . When considering a function  $\tau = h(\theta)$  of the parameter, we use the  **$\delta$ -method**. For a single parameter, If  $\tau = h(\theta)$  and  $h(\cdot)$  is differentiable with  $h'(\theta) \neq 0$ , then:

$$\frac{\hat{\tau}_n - \tau}{\text{se}(\hat{\tau}_n)} \xrightarrow{d} \mathcal{N}(0, 1), \quad (11.29)$$

where:

$$\text{se}(\hat{\tau}_n) = \left| h'(\hat{\theta}_n) \right| \cdot \text{se}(\hat{\theta}_n). \quad (11.30)$$

Thus, the asymptotic confidence interval for  $\tau$  is:

$$\left[ \hat{\tau}_n - z_{\alpha/2} \text{se}(\hat{\tau}_n), \hat{\tau}_n + z_{\alpha/2} \text{se}(\hat{\tau}_n) \right]. \quad (11.31)$$

#### Example: Bernoulli Model

Consider  $X_1, \dots, X_n \sim \text{Ber}(p)$  with  $h(p) = \log \frac{p}{1-p}$  (logit transformation).

- The **\*\*MLE\*\*** of  $p$  is  $\hat{p}_n = \bar{X}_n$ .
- The **\*\*asymptotic standard error\*\*** of  $\hat{p}_n$  is:

$$\text{se}(\hat{p}_n) = \sqrt{\frac{\hat{p}_n(1 - \hat{p}_n)}{n}}. \quad (11.32)$$

- The derivative of  $h(p)$  is:

$$h'(p) = \frac{1}{p} + \frac{1}{1-p} = \frac{1}{p(1-p)}. \quad (11.33)$$

- Applying the  $\delta$ -method:

$$\text{se}(\hat{\tau}_n) = |h'(\hat{p}_n)| \cdot \text{se}(\hat{p}_n) = \frac{1}{\hat{p}_n(1 - \hat{p}_n)} \cdot \sqrt{\frac{\hat{p}_n(1 - \hat{p}_n)}{n}}. \quad (11.34)$$

- The **\*\*asymptotic confidence interval\*\*** for  $\tau$  is:

$$\left[ \hat{\tau}_n - z_{\alpha/2} \text{se}(\hat{\tau}_n), \hat{\tau}_n + z_{\alpha/2} \text{se}(\hat{\tau}_n) \right]. \quad (11.35)$$

## 11.6 Parametric Bootstrap

For parametric models, standard errors and confidence intervals can also be estimated using the **parametric bootstrap**.

- In the **nonparametric bootstrap**, we sample  $\{X_1^*, \dots, X_n^*\}$  from the empirical distribution  $\hat{F}_n(\cdot)$ .
- In the **parametric bootstrap**, we instead sample from  $f(\cdot | \hat{\theta}_n)$ , where  $\hat{\theta}_n$  is a consistent estimator, often the MLE.

Compared to the  $\delta$ -method:

- The bootstrap is computationally simpler but requires resampling.
- The  $\delta$ -method provides a closed-form expression for the standard error.

### 11.6.1 Nonparametric Bootstrap

- **Assumption:** No assumption on the underlying true distribution.
- **Sampling:** Bootstrap samples  $\{X_1^*, \dots, X_n^*\}$  are drawn from the empirical distribution  $\hat{F}_n(\cdot)$ .
- **Estimator:** The parameter is estimated using:

$$\hat{\theta}_n = g(\hat{F}_n)$$

- **Advantages:**
  - Fully data-driven and flexible.
  - Works well when the true distribution is unknown.
- **Limitations:**
  - Fails when the sample size is too small.
  - Struggles if data are not i.i.d.
  - Can perform poorly if the parameter function  $\theta = g(F_X)$  is not smooth.

### 11.6.2 Parametric Bootstrap

- **Assumption:** Assumes a specific parametric model  $F_\theta(\cdot)$ .
- **Sampling:** Bootstrap samples  $\{X_1^*, \dots, X_n^*\}$  are drawn from the estimated parametric distribution  $F_{\hat{\theta}_n}(\cdot)$ .
- **Estimator:** The parameter is estimated as:

$$\hat{\theta}_n = g(X_1, \dots, X_n)$$

- **Advantages:**
  - More efficient for small sample sizes when the model is correct.
  - Computationally lighter compared to the nonparametric bootstrap.
- **Limitations:**
  - Fails when the assumed parametric model is incorrect.

### 11.6.3 Importance of Model Assumptions

- If  $\theta$  is defined only in terms of a parametric model (e.g., via MLE), the parametric bootstrap is the logical choice.
- Resampling from  $F_{\hat{\theta}_n}(\cdot)$  is **not** analogous to finite population sampling but rather to simulating from a statistical model.

## 11.7 Model Criticism and Selection

### 11.7.1 How Do We Choose the Right Model?

- Check the assumptions informally (e.g., histogram inspection).
- Use formal model selection criteria:
  - Akaike Information Criterion (AIC).
  - Bayesian Information Criterion (BIC).
  - Cross-validation.

### 11.7.2 Example: Modeling Failure Times

Consider data  $\{X_1, \dots, X_n\}$  representing times-to-failure of hard drives. Popular models include:

- $\mathcal{M}_1$ : Exponential distribution  $f_X(x|\lambda) = \lambda e^{-\lambda x}$ .
- $\mathcal{M}_2$ : Gamma distribution  $f_X(x|\alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$ .
- $\mathcal{M}_3$ : Log-normal distribution where  $\log X \sim \mathcal{N}(\mu, \sigma^2)$ .

We distinguish two goals:

- **Best predictive model**: AIC and cross-validation.
- **Finding the "true" model**: BIC.

## 11.8 Akaike Information Criterion (AIC)

### 11.8.1 AIC: Definition and Motivation

Given a family of models  $\mathcal{M} = \{M_1, \dots, M_k\}$ , where:

$$M_j = \{f_j(x|\theta_j) : \theta_j \in \Theta_j\}, \quad j \in \{1, \dots, k\}, \quad (11.36)$$

we define the AIC for model  $M_j$  as:

$$\text{AIC}_j = 2\ell(\hat{\theta}_j) - 2d_j, \quad (11.37)$$

where:

- $\ell(\hat{\theta}_j)$  is the log-likelihood evaluated at the MLE.
- $d_j = \dim(\Theta_j)$  is the number of parameters in the model.

The model with the highest AIC is preferred.

### 11.8.2 Justification via Kullback-Leibler Divergence

The AIC minimizes the expected Kullback-Leibler divergence:

$$D_{KL}(f_X \parallel f_j) = \int f_X(x) \log \frac{f_X(x)}{f_j(x|\hat{\theta}_j)} dx. \quad (11.38)$$

This provides a measure of how well model  $M_j$  approximates the true distribution.

## 11.9 Cross-Validation (CV)

### 11.9.1 Basic Idea

- The data are split into a training set and a test set.
- The models are fit on the training set and used to predict the test set.
- Performance is averaged over multiple splits (e.g.,  $k$ -fold cross-validation).

### 11.9.2 Single-Split Cross-Validation

For models  $\mathcal{M} = \{M_1, \dots, M_k\}$ :

- Split data into:
  - Training set:  $\{X_1, \dots, X_n\}$ .
  - Test set:  $\{X_1^*, \dots, X_n^*\}$ .

- Compute:

$$CV_j = \frac{1}{n} \sum_{i=1}^n \log f_j(X_i^* | \hat{\theta}_j). \quad (11.39)$$

- Select model with the highest CV score.

## 11.10 Bayesian Information Criterion (BIC)

### 11.10.1 Definition and Interpretation

The BIC is defined as:

$$BIC_j = \ell(\hat{\theta}_j) - \frac{d_j}{2} \log n. \quad (11.40)$$

Compared to AIC:

- BIC penalizes model complexity more strongly.
- It approximates Bayes' factor under certain conditions.

### 11.10.2 Bayesian Justification

Given prior probabilities  $\lambda_j$  over models and priors  $p_j(\theta_j)$  over parameters, Bayes' rule gives:

$$\Pr(M_j|X) \propto \lambda_j \int L(\theta_j|X) p_j(\theta_j) d\theta_j. \quad (11.41)$$

Schwarz (1978) showed that:

$$\log \Pr(M_j|X) \approx \text{BIC}_j. \quad (11.42)$$

## 11.11 Model Selection

**Model selection** refers to the process of choosing the best statistical model among a set of candidate models. Given a dataset  $X_1, X_2, \dots, X_n$ , we often assume that it is generated by an unknown true distribution  $F_X(\cdot)$ . The objective is to determine the best approximation among candidate models  $\mathcal{M} = \{M_1, M_2, \dots, M_k\}$ .

- **Informal approach:** Visualizing data (e.g., histograms, Q-Q plots) to check assumptions.
- **Formal approach:** Using statistical criteria such as:
  - Akaike Information Criterion (AIC)
  - Bayesian Information Criterion (BIC)
  - Cross-validation methods

### 11.11.1 Akaike Information Criterion (AIC)

Given a parametric model family  $\mathcal{M} = \{M_1, M_2, \dots, M_k\}$ , we define:

- Let  $\hat{\theta}_j$  be the Maximum Likelihood Estimate (MLE) for model  $M_j$ .
- Define the estimated density:

$$\hat{f}_j(x) = f_j(x | \hat{\theta}_j)$$

- Model selection minimizes the Kullback-Leibler divergence:

$$L(f_X, \hat{f}_j) = D_{KL}(f_X \parallel \hat{f}_j) = \int_X f_X(x) \log \frac{f_X(x)}{\hat{f}_j(x)} dx.$$

Since the first term in  $D_{KL}$  is independent of the model, we minimize:

$$\max_j -H(f_X, \hat{f}_j) = \max_j \int_X f_X(x) \log \hat{f}_j(x) dx.$$

## Bias Correction and AIC Formula

Akaike showed that the MLE log-likelihood has a downward bias, approximately:

$$\hat{R}_j = \frac{1}{n} \sum_{i=1}^n \log f_j(X_i | \hat{\theta}_j).$$

The corrected version is:

$$AIC_j = 2n\hat{R}_j - 2d_j = 2\ell(\hat{\theta}_j) - 2d_j,$$

where  $d_j$  is the number of parameters in model  $M_j$ .

### 11.11.2 Cross-Validation for Model Selection

#### Basic Cross-Validation

- Split data into a training set and a test set.
- Fit models using the training set and evaluate them on the test set.
- The best model minimizes the average test error.

For a model  $M_j$  with parameters  $\hat{\theta}_j$ , we define the cross-validation risk:

$$CV_j = \frac{1}{n} \sum_{i=1}^n \log f_j(X_i^* | \hat{\theta}_j),$$

where  $X_i^*$  are held-out test observations.

#### K-Fold Cross-Validation

- Partition data into  $K$  subsets.
- Train on  $K - 1$  subsets and test on the remaining one.
- Repeat for all  $K$  folds and average the results.

### 11.11.3 Bayesian Information Criterion (BIC)

BIC is similar to AIC but with a stronger penalty for model complexity:

$$BIC_j = \ell(\hat{\theta}_j) - \frac{\log n}{2} d_j.$$

Key properties:

- BIC penalizes complex models more than AIC.
- If the true model is in  $\mathcal{M}$ , BIC is more likely to select it.
- AIC is better for predictive accuracy, while BIC is better for model identification.

# Chapter 12

## Hypothesis Testing

Hypothesis testing is a statistical method used to make decisions about a population parameter based on sample data. It involves:

- Formulating a null hypothesis ( $H_0$ ) and an alternative hypothesis ( $H_1$ ).
- Collecting and analyzing sample data.
- \*Using a test statistic to determine whether the observed data provide sufficient evidence to reject  $H_0$ .
- Making a decision based on a predefined significance level ( $\alpha$ ).

### 12.1 Formulating Hypotheses

- **Null Hypothesis** ( $H_0$ ): Represents the status quo or no effect.
- **Alternative Hypothesis** ( $H_1$ ): Represents a deviation from the null hypothesis.

When making decisions based on statistical tests, two types of errors can occur:

|                                  | <b>Retain <math>H_0</math></b> | <b>Reject <math>H_0</math></b> |
|----------------------------------|--------------------------------|--------------------------------|
| <b><math>H_0</math> is True</b>  | Correct Decision               | Type I Error ( $\alpha$ )      |
| <b><math>H_0</math> is False</b> | Type II Error ( $\beta$ )      | Correct Decision               |

- **Type I Error** ( $\alpha$ ): Rejecting  $H_0$  when it is actually true (False Positive).
- **Type II Error** ( $\beta$ ): Failing to reject  $H_0$  when it is actually false (False Negative).
- The probability of correctly rejecting  $H_0$  when it is false is called the **Power of the Test**:  $1 - \beta$ .



## 12.2 Mechanics of a Hypothesis Test

### 12.2.1 Test Statistic

A test statistic is a function of the sample data that measures the deviation from  $H_0$ . Common examples include:

- **Z-statistic:** Used when the population variance is known.

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$$

- **T-statistic:** Used when the population variance is unknown.

$$T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$$

### 12.2.2 Rejection Region

The rejection region  $R_\alpha$  is defined as:

$$R_\alpha = \{x_1, x_2, \dots, x_n \mid T(x_1, x_2, \dots, x_n) > k_\alpha\}$$

where  $k_\alpha$  is a critical value that depends on the significance level  $\alpha$ .

### 12.2.3 Significance Level and p-value

- The **significance level**  $\alpha$  is the probability of making a Type I error (e.g., 0.05 or 5%).
- The **p-value** is the probability of obtaining a test statistic as extreme as the observed one, under  $H_0$ .
- If the p-value is less than  $\alpha$ , we reject  $H_0$ .

## 12.3 Power of a Test

**Definition:** The power of a test is the probability of correctly rejecting  $H_0$  when it is false:

$$\text{Power} = P(\text{Reject } H_0 \mid H_0 \text{ is false}) = 1 - \beta.$$

A high power means the test is effective at detecting differences when they exist.

### Factors Affecting Power

- **Sample Size ( $n$ ):** Larger samples lead to higher power.
- **Significance Level ( $\alpha$ ):** Increasing  $\alpha$  increases power but also increases Type I errors.
- **Effect Size:** Larger differences between  $H_0$  and  $H_1$  increase power.
- **Variability of Data:** Lower variance increases power.

## 12.4 Wald Test

### 12.4.1 Hypothesis Setup

Consider a parameter  $\theta \in \Theta$  where  $\Theta \subseteq \mathbb{R}$ .

- A **simple hypothesis** tests whether  $\theta = \theta_0$ .
- A **composite hypothesis** tests whether  $\theta \neq \theta_0$ .

A hypothesis test of the form:

$$H_0 : \theta = \theta_0 \quad \text{vs} \quad H_1 : \theta \neq \theta_0$$

is called a **two-sided test**. If the test is:

$$H_0 : \theta \leq \theta_0 \quad \text{vs} \quad H_1 : \theta > \theta_0$$

or

$$H_0 : \theta \geq \theta_0 \quad \text{vs} \quad H_1 : \theta < \theta_0$$

it is called a **one-sided test**.

## 12.5 Definition of the Wald Test

The Wald test is a widely used statistical test to assess the significance of estimated parameters in a model. It is particularly useful in **large-sample approximations**.

The Wald test is based on the assumption that the estimator  $\hat{\theta}$  is asymptotically normally distributed:

$$W = \frac{\hat{\theta} - \theta_0}{\text{se}(\hat{\theta})} \sim N(0, 1) \quad \text{under } H_0.$$

where  $\text{se}(\hat{\theta})$  is the estimated standard error of  $\hat{\theta}$ . The test decision rule is:

$$\text{Reject } H_0 \text{ if } |W| > z_{\alpha/2}.$$

where  $z_{\alpha/2}$  is the critical value from the standard normal distribution.

### 12.5.1 Alternative Version

An alternative version of the Wald test statistic uses the standard error under  $H_0$ :

$$W_0 = \frac{\hat{\theta} - \theta_0}{\text{se}_0}.$$

where  $\text{se}_0$  is computed assuming  $\theta = \theta_0$ .

### 12.5.2 Type I Error Probability (Size)

Under the null hypothesis  $H_0$ , we assume:

$$W \sim N(0, 1).$$

Thus, the probability of a **Type I error** (false positive) is:

$$P(\text{Reject } H_0 | H_0 \text{ is true}) = P(|W| > z_{\alpha/2}) = \alpha.$$

### 12.5.3 Power of the Wald Test

If the true value of  $\theta$  is  $\theta^* \neq \theta_0$ , then the probability of correctly rejecting  $H_0$  (power) is:

$$P(\text{Reject } H_0 | H_0 \text{ is false}) = 1 - \Phi\left(\frac{\theta_0 - \theta^*}{\text{se}(\hat{\theta})} + z_{\alpha/2}\right) + \Phi\left(\frac{\theta_0 - \theta^*}{\text{se}(\hat{\theta})} - z_{\alpha/2}\right).$$

## 12.6 Definition and Interpretation of p-Values

Hypothesis tests traditionally make decisions by comparing a **test statistic** with a **critical threshold**  $k_\alpha$ . Instead, we can **move from the test-statistic scale to the probability scale**, this leads to the concept of the **\*\*p-value\*\***.

- A p-value is the smallest significance level  $\alpha$  at which the test still rejects  $H_0$ .
- Formally, for a test statistic  $T(X_n)$  with rejection region  $R_\alpha$ :

$$\text{p-value} = \inf \{ \alpha \in (0, 1) : T(X_n) \in R_\alpha \}.$$

- It quantifies evidence against  $H_0$ :
  - **Small p-value**  $\rightarrow$  strong evidence against  $H_0$ .
  - **Large p-value**  $\rightarrow$  weak or no evidence against  $H_0$ .

| p-Value Range        | Interpretation                      |
|----------------------|-------------------------------------|
| $p \leq 0.01$        | Very strong evidence against $H_0$  |
| $0.01 < p \leq 0.05$ | Strong evidence against $H_0$       |
| $0.05 < p \leq 0.1$  | Weak evidence against $H_0$         |
| $p > 0.1$            | Little or no evidence against $H_0$ |

## 12.7 Multiple Testing

In some situations, we conduct many hypothesis tests simultaneously (e.g. neuroimaging, machine learning). The probability of a false rejection (Type I error) for one test is  $\alpha$ . However, when testing **\*\*multiple hypotheses\*\***, the probability of at least one false rejection is much higher! Suppose we test  $m$  null hypotheses:  $\{H_0^{(1)}, H_0^{(2)}, \dots, H_0^{(m)}\}$ . If each test is conducted at level  $\alpha$ , the probability of at least one Type I error increases as  $m$  grows.

Example: Imagine flipping  $m$  coins, each having probability  $\alpha$  of landing on heads. The probability of getting at least one head (false positive) is:

$$1 - (1 - \alpha)^m \approx 1 - e^{-m\alpha}$$

When  $m$  is large, this probability approaches 1! So we are almost certain to make at least one false discovery, even if all null hypotheses are true.

## 12.8 Mechanics of Multiple Testing

- We have  $m$  hypothesis tests, out of which  $m_0$  are truly null. We Define:

$$M_0 = \{k \in \{1, \dots, m\} : H_0^{(k)} \text{ is true}\}.$$

- For each test, we compute a p-value  $p^{(k)}$ . The smaller  $p^{(k)}$ , the more likely the corresponding null hypothesis is false. We arrange p-values in ascending order and decide on a threshold  $t$  to determine which hypotheses to reject.

### 12.8.1 Error Control Strategies

#### Family-Wise Error Rate (FWER)

Controls the probability of **at least one** false positive:

$$\text{FWER}(t) = P(RF(t) \geq 1).$$

This is a strict criterion ensuring that false discoveries are very rare.

**Bonferroni Correction:** A simple method to control FWER is to divide  $\alpha$  by the number of tests:

$$p^{(k)} \leq t_{\text{Bon}} = t = \frac{\alpha}{m}.$$

The probability of making at least one false rejection (Type I error) is:

$$P(RF(t) \geq 1) = P(\text{Making a false rejection}) = P\left(\bigcup_{k \in M_0} p^{(k)} \leq \frac{\alpha}{m}\right).$$

Using the union bound, we get:

$$P(RF(t) \geq 1) \leq \sum_{k \in M_0} P(p^{(k)} \leq \frac{\alpha}{m}) = m_0 \cdot \frac{\alpha}{m} \leq \alpha.$$

This guarantees control over the overall error rate but can lead to low power.

### 12.8.2 False Discovery Rate (FDR)

Instead of controlling the probability of any false discovery, FDR controls the **expected proportion** of false discoveries:

$$\text{FDR}(t) = E\left[\frac{RF(t)}{R(t)}\right].$$

This is a more relaxed criterion than FWER, allowing for some false positives in exchange for greater detection power.

**Benjamini–Hochberg (BH) Procedure:**

1. Order the p-values:  $p^{(1)} \leq p^{(2)} \leq \dots \leq p^{(m)}$ .
2. Find the largest  $k$  such that:

$$p^{(k)} \leq \frac{k}{m}\alpha.$$

3. Set the threshold:

$$t_{\text{BH}} = p^{(k)}.$$

4. **Decision Rule:** Reject  $H_0^{(k)}$  for all  $k \leq j$  where  $p^{(k)} \leq t_{\text{BH}}$ .

## 12.9 Likelihood Ratio Test (LRT)

Consider a parametric model  $\mathcal{F} = f(x | \theta) : \theta \in \Theta$  and testing:

$$H_0 : \theta \in \Theta_0 \quad \text{vs.} \quad H_1 : \theta \notin \Theta_0.$$

Given an i.i.d. sample  $X_n = X_1, \dots, X_n$ , the likelihood ratio test rejects  $H_0$  if  $\lambda(X_n) \leq c$ , where:

$$\lambda(X_n) = \frac{\sup_{\theta \in \Theta_0} L(\theta | X_n)}{\sup_{\theta \in \Theta} L(\theta | X_n)} = \frac{L(\hat{\theta}_0 | X_n)}{L(\hat{\theta} | X_n)},$$

where  $\hat{\theta}_0$  is the MLE constrained to  $\Theta_0$ .

### 12.9.1 Asymptotic Approximation

For scalar  $\theta$ , under  $H_0$ :

$$T_n(X_n) = -2 \log \lambda(X_n) \xrightarrow{d} \chi_1^2.$$

For vector-valued  $\theta \in \mathbb{R}^p$ , under suitable conditions:

$$T_n(X_n) = -2 \log \lambda(X_n) \xrightarrow{d} \chi_\nu^2,$$

where  $\nu = \dim(\Theta) - \dim(\Theta_0)$ .

## 12.10 Goodness-of-Fit Tests

### 12.10.1 Chi-Square Test

Divide the range of the data into  $m$  disjoint intervals  $I_1, \dots, I_m$ . Compute:

$$p_j(\theta) = \int_{I_j} f(x | \theta) dx.$$

Define the test statistic:

$$Q = \sum_{j=1}^m \frac{(N_j - np_j(\hat{\theta}))^2}{np_j(\hat{\theta})} \xrightarrow{d} \chi_{m-(p+1)}^2.$$

### 12.10.2 Kolmogorov-Smirnov Test

Define the empirical distribution function:

$$F_n(x) = \frac{1}{n} \sum_{i=1}^n I_{(-\infty, x]}(X_i).$$

The KS test statistic is:

$$T_n = \sup_x |F_n(x) - F(x)|.$$

The test is based on the asymptotic distribution of  $T_n$ .

## 12.11 Two-Sample Permutation Test

Given two independent samples  $X_1, \dots, X_n \sim F$  and  $Y_1, \dots, Y_m \sim G$ , test:

$$H_0 : F = G \quad \text{vs.} \quad H_1 : F \neq G.$$

Define labels  $L = 1, \dots, 1, 2, \dots, 2$  and test statistic:

$$T = |\bar{X} - \bar{Y}|.$$

Compute the permutation p-value:

$$p = \frac{1}{N!} \sum_{\pi} I(T_{\pi} \geq T),$$

where  $T_{\pi}$  is the test statistic under permutation  $\pi$ .